



传统基于异质网络的中药毒性预测

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文章信息

关键词: 传统
Chinese medicine 草药
toxicity 毒性prediction 异质
network 对比learning

摘要

clinical usage of Traditional Chinese Medicines (TCMs) gains increasing international attention for their distinct therapeutic effects. Precisely understanding the herb (TCM) toxicity and identifying toxic and effective herb ingredients are crucial for their safe use. Traditional wet-lab based pipelines for assessing herb toxicity are complex and time-consuming. Given the large volume toxicity data of ingredients, building computational models for efficient ingredient-based toxicity prediction and evaluation is promising, but often fails to effectively predict the toxicity of herbs due to the complexity herbs and toxic mechanisms. To address these challenges, we propose a heterogeneous network based approach (HerbToxNet) for predicting herb toxicity. HerbToxNet first constructs a heterogeneous network composed with herbs, ingredients and molecular targets, and leverages a heterogeneous graph attention network to learn the representation of herbs. Meanwhile, it performs contrastive learning with dynamic coefficient to refine the representation by pulling close the herbs with shared toxic labels, while pushing away the others. Furthermore, we introduce a weighted label fusion strategy that uses toxic labels of similar herbs to augment the predicted labels of this herb. HerbToxNet outperforms competitive methods and finds out novel potential toxicities. It uses Multilayer Perceptron (MLP) on the herb representation to predict the toxic labels of this herb. HerbToxNet outperforms competitive methods and finds out novel potential toxicities. with 96% toxicity labels confirmed for canonical herbs. It can mine related toxic ingredients and targets in an interpretable way, and dissect the molecular mechanism of herb toxicity with authenticity.

1. 介绍

草药 (Herb) 在世界各地的传统医疗实践中有着悠久的历史, 尤其是在亚洲, 它几世纪以来一直是医疗保健的重要基石 (Hu 等, 2025)。草药由多种生物活性成分组成, 如生物碱、黄酮类和萜类, 它们共同促成了草药的治疗效果。尽管草药被认为是安全和天然的, 但许多草药已与不良反应相关, 包括肝毒性、肾毒性、心脏毒性等 (Jiang 等, 2010)。这些毒性作用可能由剂量不当、长期使用、草药相互作用或重金属和农药污染引起。此外, 由于栽培、采收和加工方法的差异导致的草药质量差异, 进一步增加了其安全性评估的复杂性 (Liu 等, 2020)。面对这些多方面的挑战, 迫切需要开发有效的

使用高效且可靠的方法系统地识别和评估草药的毒性。

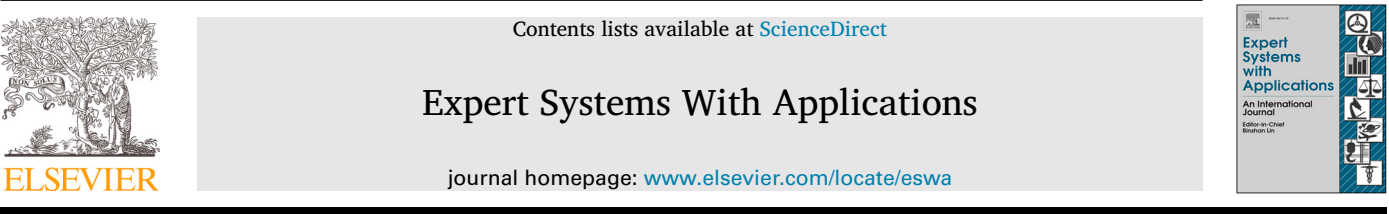
与化学药物的简单成分不同, 草药含有多种生物活性成分, 具有多靶点、多通路和多效应的相互作用, 其毒性机制更加复杂多样 (Wu 等, 2021)。因此, 草药毒性研究面临一些独特的挑战。首先, 草药毒性研究需要复杂且昂贵的实验。不同文献的数据往往存在显著的异质性和不一致性, 而临床数据分散在各机构之间, 缺乏标准化格式和共享平台。这使得获取高质量毒性数据变得困难 (Chan, 2003)。其次, 草药的多成分特性和复杂毒性机制对模型提出了更高的要求。然而, 传统的机器学习和深度学习方法难以处理草药的多模态特征 (Wang 等, 2004)。第三, 草药

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<https://doi.org/10.1016/j.eswa.2025.129969> 已收到

30 July 2025; Received in revised form 21 September 2025; Accepted 6 October 2025

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Traditional Chinese medicine toxicity prediction by heterogeneous network

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A R T I C L E I N F O

Keywords:
Traditional Chinese medicine
Herb toxicity
Toxicity prediction
Heterogeneous network
Contrastive learning

A B S T R A C T

The clinical usage of Traditional Chinese Medicines (TCMs) gains increasing international attention for their distinct therapeutic effects. Precisely understanding the herb (TCM) toxicity, and identifying toxic and effective herb ingredients are crucial for their safe use. Traditional wet-lab based pipelines for assessing herb toxicity are complex and time-consuming. Given the large volume toxicity data of ingredients, building computational models for efficient ingredient-based toxicity prediction and evaluation is promising, but often fails to effectively predict the toxicity of herbs, due to the complexity herbs and toxic mechanisms. To address these challenges, we propose a heterogeneous network based approach (HerbToxNet) for predicting herb toxicity. HerbToxNet first constructs a heterogeneous network composed with herbs, ingredients and molecular targets, and leverages a heterogeneous graph attention network to learn the representation of herbs. Meanwhile, it performs contrastive learning with dynamic coefficient to refine the representation by pulling close the herbs with shared toxic labels, while pushing away the others. Next, it uses Multilayer Perceptron (MLP) on the herb representation to predict the toxic labels of this herb, and further introduces a weighted label fusion strategy that uses toxic labels of similar herbs to augment the predicted labels of this herb. HerbToxNet outperforms competitive methods and finds out novel potential toxicities, with 96% toxicity labels confirmed for canonical herbs. It can mine related toxic ingredients and targets in an interpretable way, and dissect the molecular mechanism of herb toxicity with authenticity.

1. Introduction

Herb medicine (Herb) has a long history in traditional medical practices around the world, especially in Asia, where it has been a cornerstone of healthcare for centuries (Hu et al., 2025). Herbs consist of a variety of bioactive ingredients, such as alkaloids, flavonoids, and terpenes, which contribute to their therapeutic effects. Although they are considered as safe and natural, many herbs have been linked to adverse effects, including hepatotoxicity, nephrotoxicity, cardiotoxicity, etc. (Jiang et al., 2010). These toxic effects can be caused by improper dosing, prolonged use, herb interactions, or contamination with heavy metals and pesticides. In addition, variations in herb quality arising from differences in cultivation, harvesting, and processing methods cause further complexity into their safety evaluation (Liu et al., 2020). Given these multifaceted challenges, there is an urgent need to develop effi-

cient and reliable methodologies to systematically identify and evaluate herb toxicity.

Different from the simple composition of chemical drugs, a herb contains a variety of bioactive ingredients with multi-target, multi-pathway and multi-effect interactions, and its toxicity mechanisms are more complex and diverse (Wu et al., 2021). Therefore, the study of herb toxicity faces some distinctive challenges. First, herb toxicity studies require complex and costly experiments. Data from different literature often exhibit substantial heterogeneity and inconsistency, while clinical data are scattered across institutions, lacking standardized formats and shared platforms. This makes it difficult to obtain high-quality toxicity data (Chan, 2003). Second, the multi-ingredient nature and complex toxicity mechanisms of herbs impose higher demands on models. However, traditional machine learning and deep learning methods struggle to process the multi-modal features of herbs (Wang et al., 2004). Third, herb

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<https://doi.org/10.1016/j.eswa.2025.129969>

Received 30 July 2025; Received in revised form 21 September 2025; Accepted 6 October 2025

Available online 10 October 2025

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毒性预测通常被简化为二分类任务，而实际上它本质上应该是多标签分类任务，因为一种草药通常与不同的毒性效应相关。在我们收集的中药毒性数据集中，超过一半的草药（252种中的141种）与两个或更多毒性标签相关，这凸显了基于多标签学习的草药毒性预测的复杂性和必要性。第四，深度学习模型通常被视为“黑箱”，使其难以为毒性预测提供明确的生物学解释（Gao 等，2025）。这种可解释性的缺乏阻碍了研究人员对毒性机制的理解，从而削弱了深度模型的真实性和临床适用性。

监管机构和研究人员一直强调提高草本产品安全性和质量控制的重要性（Jit̃areanu 等，2022）。传统的草药毒性评估方法，如体内动物测试和体外细胞分析，不仅耗时且成本高，而且还引发伦理问题。此外，鉴于草药成分数量众多且作用机制多样，全面的毒性评估仍然是一项艰巨的任务。随着高通量技术的出现以及大规模生物和化学数据集的不断增加，计算方法在草药毒性预测中受到了关注。这些方法包括机器学习模型，如随机森林、支持向量机和深度学习框架，它们能够分析复杂的、多维数据以预测毒性结果（He 等，2019a；Sun 等，2019；Yang 等，2021）。通过整合化学数据

在本文中，我们通过整理现有的草药毒性研究结果，构建了一个多标签草药毒性数据集，以解决草药毒性数据分散和稀缺的问题（见第4节）。为了解决复杂草药建模的困难以及缺乏可解释性的问题，我们提出了一种基于异构网络的草药毒性预测方法（HerbToxNet）。HerbToxNet首先利用药理特性、毒理学特征以及草药之间的关联构建草药-成分-靶点异构网络，然后使用异构图注意力网络（HAN）（Wang等，2019）学习草药的表示。同时，它引入具有动态系数的对比学习，将具有相同毒性标签的草药对表示拉近，而将具有不同毒性标签的草药对表示拉远。接着，它在草药表示上使用多层感知机（MLP）预测初始毒性标签。

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(i) 我们通过整理分散在不同草药数据库和已发表文献中的现有草药毒性研究结果，构建并共享了一个草药多标签毒性数据集。

(ii) 我们提出了一个统一框架 HerbToxNet，通过异质网络对药材、成分和靶点之间的多层次关系进行建模，该框架能够对与毒性相关的机制进行透明分析，并增强可解释性。

(iii) HerbToxNet 综合了异构网络建模、对比学习和加权标签融合，以解决毒性数据稀缺的问题，并以合理且真实的方式实现毒性预测。

(iv) 大量实验证明，HerbToxNet 的表现优于竞争方法，能够识别经典草药的潜在毒性（96% 已确认），并且可以从可解释的角度直接揭示与毒性相关的成分和靶点。

本文的其余部分安排如下。第2节讨论相关工作，第3节详细介绍所提出的HerbToxNet框架。第4节展示实验设置、结果及分析。第5节总结我们的工作并概述未来的努力方向。

2. 相关工作

现有的草药毒性计算方法通常预测一些主要的毒性类型，如肝毒性、肾毒性、心脏毒性等，并将其建模为二元分类任务（所关注的毒性是否存在）。基于草药毒性的特征，发展了一系列草药毒性预测的理论和方法，可分为三类：基于定量构效关系（QSAR）的草药毒性预测、网络毒理学、草药性质理论。

在计算毒理学中，通常使用化合物的虚拟结构，通过计算方法分析、模拟、可视化或预测化学物质的毒性（Raies & Bajic, 2016）。QSAR 可以定量捕捉分子结构与生物效应之间的关系，它赋能了化合物毒性的计算预测，并广泛应用于化学药物的毒性评估（Cherkasov et al., 2014）。因此，基于 QSAR 的方法可以被用于计算预测每种草药中各个成分（类似药物）的毒性，为估算草药整体毒性提供依据（Gao et al., 2019）。He 等（2019a）收集了肝毒性草药及其成分，分析了这些成分的结构，发现生物碱和萜类化合物是主要的肝毒性成分。Yang 等（2021）使用 QSAR 评估了决明子成分的潜在毒性，并阐明了相关关系

网络毒理学是基于网络药理学研究生物医药的重要分支（Li 等，2019）。网络毒理学的主要研究目的是通过特定的毒性-靶点-成分-药物相互作用网络分析药物毒性。一般分析过程包括：药物有毒成分的鉴定、有毒成分的预测以及有毒靶点的功能分析。Wu 等（2019a）将四种机器学习方法与四种不同的指纹集结合，构建了 16 个单一分类器来预测潜在的肝毒性成分，并通过系统药理学方法探讨药物肝毒性的机制。Yu 等（2025）在镇痛风痛宁诱导的肝损伤数据集中比较了十种机器学习模型，以寻找预测肝损伤的最有效模型。虽然网络毒理学可以系统地阐明毒性机制并识别关键靶点，但这种方法

“四气五味”（亦称“四气五味”）与归经是中药药理理论框架的重要组成部分（Qiao 等，2022）。中药的“四气”分为五类：寒、热，

toxicity prediction is often simplified as a binary classification task, whereas it should inherently be a multi-label classification task, as a herb is often associated with different toxic effects. In our collected TCM toxicity dataset, more than half of the herbs (141 out of 252) are linked with two or more toxic labels, highlighting the complexity and necessity of multi-label learning based herb toxicity prediction. Fourth, deep learning models are often seen as “black boxes”, making them difficult to provide clear biological explanations for their toxicity prediction (Gao et al., 2025). This lack of interpretability hinders researchers’ understanding of toxicity mechanisms, and thus undermines the authenticity and clinical applicability of deep models.

Regulatory authorities and researchers have been emphasizing the importance of improving the safety and quality control of herb products (Jit̃areanu et al., 2022). Traditional methods for assessing herb toxicity, such as in vivo animal testing and in vitro cellular assays, are not only time-consuming and costly but also raise ethical concerns. Moreover, given the vast number of ingredients and their diverse action mechanisms, the comprehensive toxicity assessments remain a daunting task. With the advent of high-throughput technologies and the growing availability of large-scale biological and chemical datasets, computational approaches have gained attraction in herb toxicity prediction. These approaches include machine learning models, such as random forests, support vector machines, and deep learning frameworks, which can analyze complex, multidimensional data to predict toxicity outcomes (He et al., 2019a; Sun et al., 2019; Yang et al., 2021). By integrating data on chemical structures, pharmacological activities, and known toxicological profiles, these models offer a cost-effective and scalable solution for identifying potential toxicity of herbs and facilitating their safe usage (Jia et al., 2024; Wang et al., 2020; Wu et al., 2019a; Yu et al., 2025). However, existing computational approaches still encounter significant challenges arising from the heterogeneity and incompleteness of herb toxicity related data. Most current models are limited to a single type of information, such as molecular structure or pharmacological activity, and often lack the capability to effectively integrate diverse data sources that reflect the complex biological mechanisms of herbs. Furthermore, the absence of systematic frameworks capable of capturing multi-level interactions among herbs, ingredients, targets, and toxicities, restricts both the interpretability and authenticity of toxicity predictions. Given these, there is a critical need of authentic computational approaches that can systematically fuse multi-source information and address the challenges of data heterogeneity, sparsity, and interpretability in herb toxicity prediction.

In this paper, we construct a multi-label toxicity dataset of herbs by curating the results of existing herb toxicity studies to solve the problem of dispersed and scarce herb toxicity data (see Section 4). To address the difficulties in modeling complex herbs and the lack of interpretability, we propose a heterogeneous network based approach (HerbToxNet) for herb toxicity prediction. HerbToxNet first constructs a herb-ingredient-target heterogeneous network using the pharmacological properties, toxicological characteristics, and associations between herbs, and then uses Heterogeneous Graph Attention Network (HAN) (Wang et al., 2019) to learn the representations of herbs. At the same time, it introduces contrastive learning with dynamic coefficient to pull closer the representations of herb pairs with shared toxic labels while pushing apart those with different toxic labels. Next, it uses a Multilayer Perceptron (MLP) on the herb representation to predict the initial toxic labels, and further introduces a weighted label fusion strategy that retrieves similar herbs and merges their toxic labels to refine the predicted labels.

Our contributions are summarized as below:

- (i) We construct and share a multi-label toxicity dataset of herbs by curating the results of existing herb toxicity studies scattered in different herb databases and published literature.
- (ii) We propose a unified framework HerbToxNet to model multi-level relationships among herbs, ingredients, and targets via a heterogeneous network, which allows for transparent analysis of toxicity-related mechanisms and enhances interpretability.

- (iii) HerbToxNet synergizes heterogeneous network modeling, contrastive learning, and weighted label fusion to remedy scarce toxicity data and achieve toxicity prediction in a sensible and authentic way.
- (iv) Extensive experiments show that HerbToxNet outperforms competitive methods, identifies potential toxicities of canonical herbs (96% confirmed), and can directly uncover toxicity-related ingredients and targets from an interpretable view.

The rest of the paper is organized as follows. Section 2 discusses the related work and Section 3 elaborates on the proposed HerbToxNet framework. Section 4 presents the experimental setup, results and analysis. Section 5 summarizes our work and outlines future efforts.

2. Related work

Existing computational methods for herb toxicity usually predict some main types of toxicity, such as hepatotoxicity, nephrotoxicity, cardiotoxicity, etc., and model them as binary classification tasks (whether the interested toxicity is present or not). Based on the characteristics of herb toxicity, a series of theories and methods for herb toxicity prediction were developed, which can be divided into three categories: herb toxicity prediction based on quantitative structure-activity relationship (QSAR), network toxicology, herb property theory.

In computational toxicology, the virtual structure of a compound is often used to analyze, simulate, visualize, or predict the toxicity of a chemical by computational methods (Raies & Bajic, 2016). QSAR can quantitatively capture the relationship between molecular structure and biological effects, it empowers the computational prediction of compound toxicity and has been widely applied in the toxicity assessment of chemical drugs (Cherkasov et al., 2014). Therefore, QSAR-based methods can be adapted to computationally predict the toxicity of individual ingredients (alike drugs) within each herb, providing a basis for estimating the overall toxicity of herbs (Gao et al., 2019). He et al. (2019a) collected hepatotoxic herbs and ingredients, analyzed the structure of these ingredients and found that alkaloids and terpenoids were the main hepatotoxic ingredients. Yang et al. (2021) used QSAR to evaluate the potential toxicity of ingredients in Cassia seed, and clarified the relationship between structural characteristics and cytotoxicity by linear model, and found that three ingredients in Cassia seed had hepatotoxicity and eight ingredients had nephrotoxicity. Sun et al. (2019) used herb nephrotoxicity data to construct a QSAR model, and used ANN and SVM algorithms to predict nephrotoxicity. While QSAR-based approaches enable quantitative analysis of structure-toxicity relationships at the molecular level, their accuracy and applicability in herbal toxicity prediction are limited by the availability and quality of both structural and toxicological data, as well as the incomplete characterization of many ingredients.

Network toxicology, is an important branch for studying biomedicine based on network pharmacology (Li et al., 2019). The main research purpose of network toxicology is to analyze herb toxicity through a specific toxicity-target-ingredient-herb interaction network. The general analysis process includes: identification of toxic ingredients of herb, prediction of toxic ingredients and functional analysis of toxic targets. Wu et al. (2019a) combined four machine learning methods with four different fingerprint sets to construct 16 single classifiers to predict potential hepatotoxic ingredients, and explored the mechanism of hepatotoxicity of herbs through systematic pharmacological methods. Yu et al. (2025) compared ten machine learning models in the Zhengqing Fengtongning-induced liver injury dataset to find the most effective model for predicting liver injury. While network toxicology can systematically elucidate toxic mechanisms and identify key targets, this kind of methods cannot directly quantify herb-level toxicity and are limited by incomplete herb-ingredient-target associations, suffering biased or incomplete predictions.

The “Four Properties and Five Flavors” (also known as the “Four Qi and Five Tastes”) and meridian tropism constitute critical components of the theoretical framework of herb pharmacology (Qiao et al., 2022). The “Four Properties” of herb are categorized into five types: cold, hot,

温、凉和平。“五味”分为七类：酸、苦、甘、辛、咸、淡、涩。经络归经是指将药材归入十二经：肺、大肠、胃、脾、心、小肠、膀胱、肾、心包、三焦、胆和肝。Wang 等（2020）建立了一种随机游走网络，它可以在传统人工神经网络基础上有效解释处方信息，以预测草药处方的副作用。Jia 等（2024）通过将草药毒性与电子电离质谱数据的高级分析描述符相关联，并结合可解释的机器学习模型，进行毒性预测。此外，一些研究表明，草药效能可用于分析草药毒性（Yang 等，2015；Zhu 等，2025）。尽管药性理论提供了基于传统知识的描述性见解，但其具有定性和主观性特点。

为了解决上述挑战，我们构建了一个异质图来建模草药、成分和靶点之间的复杂关系。通过利用异质图注意力网络、带有动态系数的对比学习以及加权标签融合，我们的HerbToxNet能够处理不完整数据并融合多层特征，从而有效预测草药的毒性标签。

3. 拟议的方法

HerbToxNet 的框架如图 1 所示。HerbToxNet 包括四个主要模块：(i) 异构图构建，根据草药的药理特性、功效和关联信息构建草药-成分-靶点异构图；(ii) 草药表示学习，采用异构图注意力网络（HAN）（Wang 等人，2019）结合选定的元路径提取有效的草药表示；(iii) 通过动态系数的对比学习，通过拉近具有相同特征的草药来优化学习到的表示；

有害标签并将具有不同标签的人分开；以及 (iv) 基于加权标签融合的毒性预测，根据目标草药的优化表示，融合来自多层感知器（MLP）的预测和来自相似草药的有害标签。这些模块按顺序组织，使 HerbToxNet 能够有效整合异构信息，学习稳健的表示，并实现毒性预测。

3.1. 异构图构建

草药的药理特性和功效数据包含对毒性预测有用的信息，但由于这些数据是基于传统文献和经验知识收集的，因此通常不完整或不精确。因此，现有基于草药药理特性和功效的方法往往难以充分发挥其潜力。实际上，一种草药包含多种成分，其协同作用使得毒性复杂，单靠药理特性数据难以描述和捕捉其毒性（Caesar & Cech, 2019; Flora 等，2013）。同样，网络毒理学也面临挑战，因为成分的复杂性使得现有方法难以准确预测其毒性作用。此外，这些模型往往缺乏可解释性，阻碍了对生物机制的理解，并降低了其在实际应用中的可靠性（He 等，2019a,b）。

为了解决上述问题，我们将异构草药-成分-靶点图定义为 $G = (\mathcal{V}, \mathcal{E})$ ，通过联合建模成分节点和靶点节点及其相互作用，编码草药的多成分和多靶点特性。节点集 $\mathcal{V} = \mathcal{V}_h \cup \mathcal{V}_c \cup \mathcal{V}_t$ 是草药、成分和靶点节点的并集，边集 $\mathcal{E} = \mathcal{E}_{hc} \cup \mathcal{E}_{ht} \cup \mathcal{E}_{ct} \cup \mathcal{E}_{hh}$ 是草药-成分、草药-靶点、成分-靶点以及草药-草药边的并集。我们的目标是将每种草药的药理特性和功效整合到其节点表示中，从而将领域知识与异构图的结构信息结合起来。第 i 种草药的药理特性和功效分别表示为 $\mathbf{p}_i \in \{0, 1\}^{L_p}$ 和 $\mathbf{e}_i \in \{0, 1\}^{L_e}$ 。

warm, cool, and neutral. The “Five Flavors” are classified into seven types: sour, bitter, sweet, pungent, salty, bland, and astringent. Meridian tropism refers to classifying herbs into 12 channels: lung, large intestine, stomach, spleen, heart, small intestine, bladder, kidney, pericardium, triple burner, gallbladder, and liver. Wang et al. (2020) established a random walk network that can effectively interpret prescription information on the basis of traditional ANN to predict side effects of herb prescriptions. Jia et al. (2024) made toxicity predictions by correlating herb toxicity with advanced analytical descriptors of electron ionization mass spectrometry data, combined with interpretable machine learning models. In addition, some studies have shown that the efficacy of herbs can be used to analyze herb toxicity (Yang et al., 2015; Zhu et al., 2025). While herb property theory offers descriptive insights rooted in traditional knowledge, its qualitative and subjective nature, distinct from the standardized representations used for chemical drugs, results in ambiguous data and limited predictive performance even when combined with advanced machine learning methods.

To address the above challenges, we construct a heterogeneous graph to model the complex relationships between herbs, ingredient and targets. By leveraging Heterogeneous Graph Attention Networks, contrastive learning with dynamic coefficient, and weighted label fusion, our HerbToxNet can handle incomplete data and fuse multi-level features, thereby effectively predicting the toxic labels of herbs.

3. The proposed methodology

The framework of HerbToxNet is illustrated in Fig. 1. HerbToxNet comprises four main modules: (i) Heterogeneous graph construction builds a herb-ingredient-target heterogeneous graph based on herb pharmacological properties, efficacy, associations; (ii) Herb representation learning employs a Heterogeneous Graph Attention Network (HAN) (Wang et al., 2019) with selected meta-paths to extract informative herb representations; (iii) Contrastive learning with dynamic coefficient optimizes the learned representations by pulling together herbs with shared

toxic labels and pushing apart those with dissimilar labels; and (iv) Toxicity prediction based on weighted label fusion merges the predictions from an MLP and toxic labels from similar herbs of the target herb based on their optimized representations. These modules are organized sequentially, enabling HerbToxNet to effectively integrate heterogeneous information, learn robust representations, and achieve toxicity prediction.

3.1. Heterogeneous graph construction

Herb pharmacological property and efficacy data contain useful information for toxicity prediction, but they are often incomplete or imprecise, as they are collected based on traditional literature and empirical knowledge. As such, existing methods built on herb pharmacological property and efficacy often struggle to exploit their potential. In fact, a herb has multiple ingredients with synergistic effects, making its toxicity complex and difficult to describe and capture with pharmacological property data alone (Caesar & Cech, 2019; Flora et al., 2013). Similarly, network toxicology faces challenges, as the complexity of ingredients makes it difficult for existing methods to accurately predict their toxic effects. Furthermore, these models often lack interpretability, hindering the understanding of biological mechanisms and reducing their reliability in practical applications (He et al., 2019a,b).

To address the above issues, we define the heterogeneous herb-ingredient-target graph as $G = (\mathcal{V}, \mathcal{E})$, which encode the multi-ingredient and multi-target nature of herbs by jointly modeling ingredient and target nodes, as well as their interactions. The node set $\mathcal{V} = \mathcal{V}_h \cup \mathcal{V}_c \cup \mathcal{V}_t$ is the union of herb, ingredient, and target nodes, and the edge set $\mathcal{E} = \mathcal{E}_{hc} \cup \mathcal{E}_{ht} \cup \mathcal{E}_{ct} \cup \mathcal{E}_{hh}$ is the union of herb-ingredient, herb-target, ingredient-target, and herb-herb edges. We aim to integrate both the pharmacological properties and efficacy of each herb into its node representation, thereby combining domain knowledge with structural information of the heterogeneous graph. The pharmacological properties and efficacy of i th herb are denoted as $\mathbf{p}_i \in \{0, 1\}^{L_p}$ and $\mathbf{e}_i \in \{0, 1\}^{L_e}$,

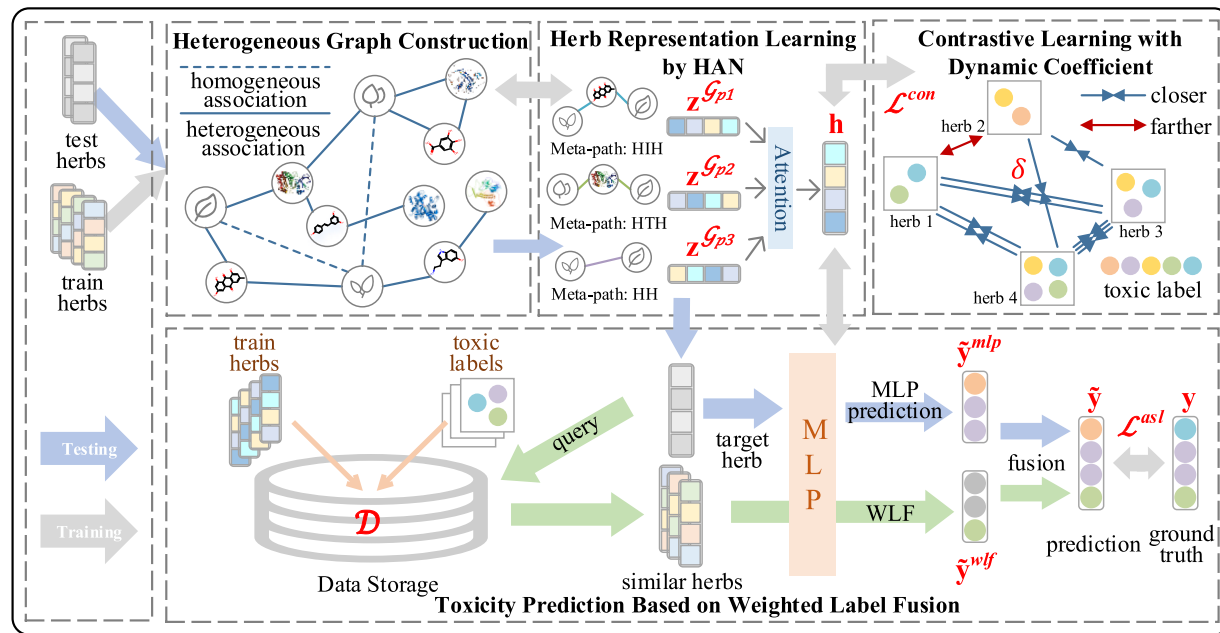


Fig. 1. The overview of HerbToxNet framework. HerbToxNet first constructs a herb-ingredient-target heterogeneous graph and selects three types of meta-paths: herb-ingredient-herb (HH), herb-target-herb (HTH) and herb-herb (HH) to build the meta-path matrix. Then it utilizes the attention module of heterogeneous graph neural network to synthesize features ($\mathbf{z}^{Gp_1}$, $\mathbf{z}^{Gp_2}$, $\mathbf{z}^{Gp_3}$) under different meta-paths to obtain the representation ($\mathbf{h}$) of the herb. Next, it introduces a contrastive learning loss ($\mathcal{L}^{con}$) with a dynamic coefficient δ to pull together the representations of herbs with shared toxic labels, while pushing apart those without shared labels thereby optimizing herb representations. After that, it predicts the toxic labels in two ways: an MLP classifier to generate $\tilde{\mathbf{y}}^{mlp}$ and a weighted label fusion strategy to aggregate labels $\tilde{\mathbf{y}}^{wlf}$ from similar herbs in the training set $\mathcal{D}$ and then merges $\tilde{\mathbf{y}}^{mlp}$ and $\tilde{\mathbf{y}}^{wlf}$ into $\tilde{\mathbf{y}}$. Finally, HerbToxNet defines asymmetric classification loss ($\mathcal{L}^{asl}$) between $\tilde{\mathbf{y}}$ and the ground-truth labels $\mathbf{y}$ to optimize the overall framework.

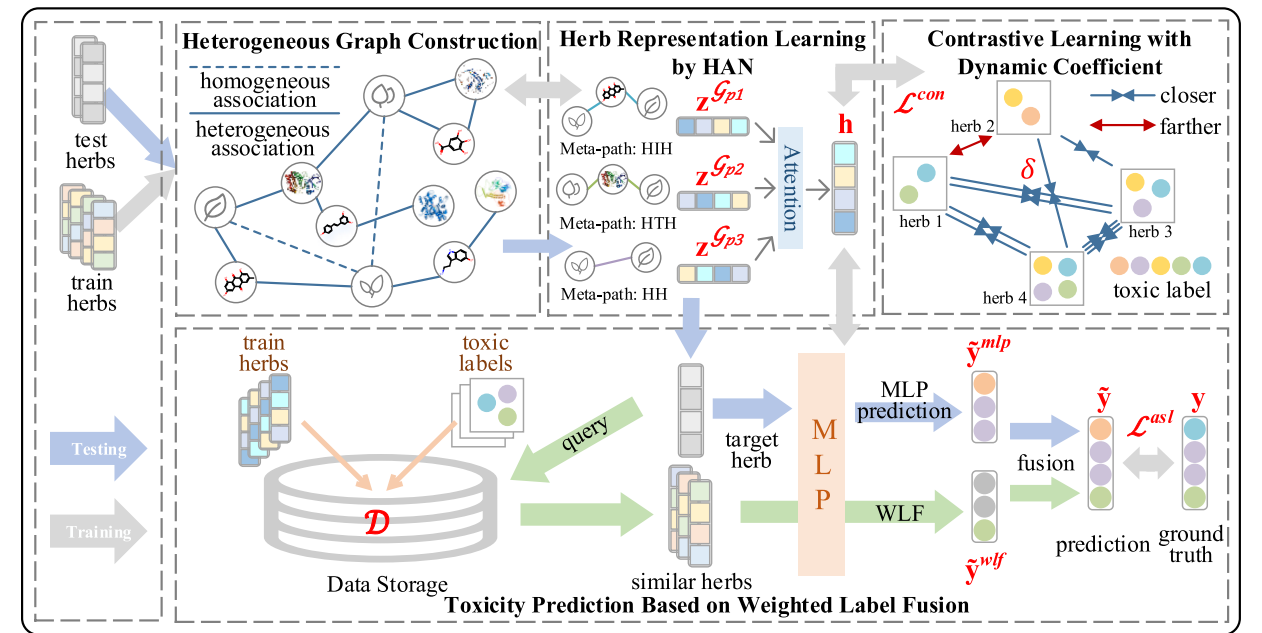


Fig. 1. The overview of HerbToxNet framework. HerbToxNet first constructs a herb-ingredient-target heterogeneous graph, and selects three types of meta-paths: herb-ingredient-herb (HH), herb-target-herb (HTH), and herb-herb (HH) to build the meta-path matrix. Then, it utilizes the attention module of heterogeneous graph neural network to synthesize features ($\mathbf{z}^{Gp_1}$, $\mathbf{z}^{Gp_2}$, $\mathbf{z}^{Gp_3}$) under different meta-paths to obtain the representation ($\mathbf{h}$) of the herb. Next, it introduces a contrastive learning loss ($\mathcal{L}^{con}$) with a dynamic coefficient δ to pull together the representations of herbs with shared toxic labels, while pushing apart those without shared labels, thereby optimizing herb representations. After that, it predicts the toxic labels in two ways: an MLP classifier to generate $\tilde{\mathbf{y}}^{mlp}$, and a weighted label fusion strategy to aggregate labels $\tilde{\mathbf{y}}^{wlf}$ from similar herbs in the training set $\mathcal{D}$, and then merges $\tilde{\mathbf{y}}^{mlp}$ and $\tilde{\mathbf{y}}^{wlf}$ into $\tilde{\mathbf{y}}$. Finally, HerbToxNet defines asymmetric classification loss ($\mathcal{L}^{asl}$) between $\tilde{\mathbf{y}}$ and the ground-truth labels $\mathbf{y}$ to optimize the overall framework.

分别表示, 其中 L_p 和 L_e 分别代表药理特性和功效的维度。若第 i 种药材具有第 j 种药理特性, 则 $p_{ij} = 1$, 否则为 $p_{ij} = 0$ 。若第 i 种药材包含第 j 种功效, 则 $e_{ij} = 1$, 否则为 $e_{ij} = 0$ 。我们使用药材的药理特性和功效作为其初始节点特征。对于第 i 种药材, 其特征表示 $\mathbf{f}_i$ 是通过对 $\mathbf{p}_i$ 和 $\mathbf{e}_i$ 的拼接应用转换得到的:

$$\mathbf{f}_i = \text{ReLU}(\mathbf{W}_h(\mathbf{p}_i || \mathbf{e}_i) + \mathbf{b}_h) \quad (1)$$

其中, $\mathbf{W}_h$ 和 $\mathbf{b}_h$ 分别表示可训练的权重矩阵和偏置向量。|| 表示拼接操作, $\text{ReLU}(\cdot)$ 是ReLU激活函数。由于我们的目标是草药毒性预测, 并且异构图将基于预定义的元路径使用HAN表示, 因此不需要为成分节点和目标节点分配特定的初始特征。

异构图 $\mathcal{G}$ 由三种节点类型组成: 草药、成分和靶点, 这些节点通过三种类型的异构关系 (草药-成分、草药-靶点和成分-靶点) 以及一种同质关系 (草药-草药) 连接。成分或靶点之间的同质关系需要显式的相似性计算或人工整理, 这会引入冗余和繁琐的工作, 因此在我们的研究中未使用这些关系。该图结构为建模草药的结构信息和属性信息提供了统一工具, 有助于毒性预测。

3.2. 通过 HAN 的草药表示学习

在异构图 $\mathcal{G}$ 上给出一个元路径 p , 该元路径形式为 $v_1 \xrightarrow{r_1} v_2 \xrightarrow{r_2} \dots \xrightarrow{r_l} v_{l+1}$, 表示节点类型 v_1 和 v_{l+1} 通过复合关系 $r = r_1 \circ r_2 \circ \dots \circ r_l$ 连接, 其中 $\circ$ 是作用于关系 $v_i \in \mathcal{V}$ 和 $r_i \in \mathcal{E}$ 的复合运算符。换句话说, 关系 r 是由那些 $r_1, r_2, \dots, r_l$ 的 l 关系组成的。因此, 元路径能够捕捉异构图中的特定语义信息, 不同的元路径代表不同的语义内容。

例如, 草药-成分-草药 (HIH) 的元路径表明两种草药含有相同的成分, 这意味着这两种草药可能具有相似的功能和靶点关联。我们通过将异构图中草药与成分的关联矩阵与其转置矩阵相乘来基于 HIH 元路径构建矩阵。同样, 草药-靶点-草药 (HTH) 的元路径表明两种草药连接到相同的靶点。草药-草药 (HH) 的元路径表示两种草药之间的相似性。HTH 和 HH 矩阵的构建方法与 HIH 矩阵类似。

在这里, 我们将异构图 $\mathcal{G}$ 分解为多个子图, 每个子图 $\mathcal{G}_p$ 由通过元路径 p 连接的所有邻居对组成, 从而被指定为基于元路径的子图。具体而言, 对于以草药节点开始或结束的元路径集合, 我们将其表示为 $\mathcal{P} = \{p_1, p_2, \dots, p_M\}$, 其中 $M = |\mathcal{P}|$ 是考虑的元路径数量。

异构图注意力网络 (HAN) (Wang 等, 2019) 是对异构图进行图表示学习的有效工具, 能够通过分层注意力机制为不同的邻居和元路径分配不同的权重。在这一背景下, 我们利用 HAN 来学习每个元路径诱导子图内的节点表示。HAN 自适应地聚合每个子图中节点的邻居信息, 从而整合节点的拓扑结构和特征。具体来说, 对于给定的节点及其关联的子图, HAN 首先采用节点级自注意力机制来学习其邻居节点的重要性 (注意力分数)。然后, 根据邻居节点的注意力分数自适应地聚合它们的特征表示。最后, 使用语义级注意力机制将来自不同元路径的表示进行组合。

对于第 k 个基于元路径的子图 $\mathcal{G}_{pk}$, 节点对 (i, j) 的重要性 $e_{ij}^{G_{pk}}$ 可以表示如下:

$$e_{ij}^{G_{pk}} = \text{LeakyReLU}(\mathbf{a}_{G_{pk}}^T \cdot [\mathbf{f}_i^{G_{pk}} || \mathbf{f}_j^{G_{pk}}]) \quad (2)$$

其中 $\mathbf{a}_{G_{pk}}$ 是第 k 条元路径的注意力参数。为了使不同元路径的注意力系数具有可比性, 我们使用 softmax 函数对 $e_{ij}^{G_{pk}}$ 进行归一化:

$$e_{ij}^{G_{pk}'} = \frac{\exp(e_{ij}^{G_{pk}})}{\sum_{j' \in \mathcal{V}_{hi}^{G_{pk}}} \exp(e_{ij'}^{G_{pk}})} \quad (3)$$

其中, $\mathcal{V}_{hi}^{G_{pk}}$ 表示节点 i 在第 k 条元路径中的邻居节点集合。接下来, 我们使用ReLU激活函数对所有元路径实例的特征向量进行聚合, 以基于第 k 条元路径得到节点 i 的表示:

$$\mathbf{f}_i^{G_{pk}} = \text{ReLU} \left(\sum_{j \in \mathcal{V}_{hi}^{G_{pk}}} e_{ij}^{G_{pk}'} \cdot \mathbf{f}_j^{G_{pk}} \right) \quad (4)$$

在本研究中, 我们引入了一种多头注意力机制, 以增强注意力系数学习过程的稳定性, 并减轻单个注意力头的影响。具体而言, 我们独立重复注意力机制 K 次, 并将每个注意力头学习到的向量表示进行拼接。因此, 第 k 条元路径下节点 i 的表示 $\mathbf{z}_i^{G_{pk}}$ 表示为:

$$\mathbf{z}_i^{G_{pk}} = ||_{j=1}^K \mathbf{f}_{ij}^{G_{pk}} \quad (5)$$

一般来说, 异构图中的每个节点包含多种类型的语义信息, 而在单一元路径下对节点特征的聚合只能反映节点信息的一个方面。为了学习更全面的节点嵌入, 有必要整合由多条元路径表示的语义信息。从不同元路径聚合嵌入的一个简单方法是取平均值。然而, 由于元路径对异构图中节点 i 的重要性不同, 我们采用注意力机制为每条元路径模式分配权重, 然后进行聚合。这确保了包含最关键语义信息的嵌入获得最大的重视。更具体地说, 我们学习每个基于元路径的子图的重要性, 并获得 $|\mathcal{P}|$ 组节点嵌入, 每组表示对应元路径下所有节点的加权聚合:

$$w_{G_{pk}} = \frac{1}{|\mathcal{V}_h|} \sum_{i \in \mathcal{V}_h} \mathbf{q}^T \cdot \tanh(\mathbf{W}_k \cdot \mathbf{z}_i^{G_{pk}} + \mathbf{b}_k) \quad (6)$$

$\mathbf{W}_k$ 、 $\mathbf{b}_k$ 和 $\mathbf{q}_k$ 分别表示与第 k 条元路径相关的权重矩阵、偏置向量和语义级注意力向量。 $\tanh(\cdot)$ 代表 $\tanh$ 激活函数。在获得特定节点不同元路径的注意力分数后, 我们使用 softmax 函数对其进行归一化, 从而能够对不同元路径的语义信息进行有意义的比较。

$$w_{G_{pk}'} = \frac{\exp(w_{G_{pk}})}{\sum_{j=1}^{|\mathcal{P}|} \exp(w_{G_{pj}})} \quad (7)$$

最后, 我们根据各自的注意力分数整合来自不同元路径的节点嵌入。第 i 个草药节点嵌入 $\mathbf{h}_i$ 如下:

$$\mathbf{h}_i = \sum_{k=1}^{|\mathcal{P}|} w_{G_{pk}'} \cdot \mathbf{z}_i^{G_{pk}} \quad (8)$$

respectively, where L_p and L_e represent the dimension of pharmacological properties and efficacy. $p_{ij} = 1$ if i th herb possesses the j th pharmacological properties, and $p_{ij} = 0$ otherwise. $e_{ij} = 1$ if i th herb contains the j th efficacy, and $e_{ij} = 0$ otherwise. We use the pharmacological properties and efficacy of herbs as their initial node features. For the i th herb, its feature representation $\mathbf{f}_i$ is obtained by applying a transformation to the concatenation of $\mathbf{p}_i$ and $\mathbf{e}_i$ as:

$$\mathbf{f}_i = \text{ReLU}(\mathbf{W}_h(\mathbf{p}_i || \mathbf{e}_i) + \mathbf{b}_h) \quad (1)$$

where $\mathbf{W}_h$ and $\mathbf{b}_h$ denote the trainable weight matrix and bias vector, respectively. || represents the concatenation operation, and $\text{ReLU}(\cdot)$ is the ReLU activation function. Since our objective is herb toxicity prediction, and the heterogeneous graph will be represented using HAN based on predefined meta-paths. Therefore, there is no need to assign specific initial features to the ingredient and target nodes.

The heterogeneous graph $\mathcal{G}$ consists of three node types: herb, ingredient, and target, which are connected by three types of heterogeneous relations (herb-ingredient, herb-target, and ingredient-target), as well as one type of homogeneous relation (herb-herb). Homogeneous relations between ingredients or targets require explicit similarity computations or manual curation, which introduce redundancy and tedious efforts, so these relations are not used in our work. This graph structure provides a unified tool for modeling both the structure and attribute information of herbs, facilitating toxicity prediction.

3.2. Herb representation learning by HAN

Give a meta-path p on the heterogeneous graph $\mathcal{G}$, with the meta-path taking the form of $v_1 \xrightarrow{r_1} v_2 \xrightarrow{r_2} \dots \xrightarrow{r_l} v_{l+1}$, denoting that the node types v_1 and v_{l+1} are connected by a composite relationship $r = r_1 \circ r_2 \circ \dots \circ r_l$, where $\circ$ is the composite operator on the relationship, $v_i \in \mathcal{V}$ and $r_i \in \mathcal{E}$. In other words, the relationship r is composed of l relationships of those $r_1, r_2, \dots, r_l$. Thus, meta-path can capture specific semantic information within a heterogeneous graph, with different meta-paths representing different semantic contents.

For example, the meta-path of Herb-Ingredient-Herb (HIH) suggests that both herbs contain the same ingredient, implying that these two herbs may possess similar functionalities and target associations. We construct the matrix based on the HIH meta-path by multiplying the association matrix of herbs and ingredients in the heterogeneous graph with its transpose. Similarly, meta-path Herb-Target-Herb (HTH) suggests that both herbs link with the same target. The meta-path Herb-Herb (HH) indicates the similarity between two herbs. The construction methods for the HTH and HH matrices are similar to that of the HIH matrix.

Here, we decompose the heterogeneous graph $\mathcal{G}$ into multiple subgraphs, each subgraph $\mathcal{G}_p$ is composed of all neighbor pairs connected by the meta-path p , thereby designated as meta-path based subgraphs. Specifically, for the set of meta-paths that start or end with herb nodes, we denote it as $\mathcal{P} = \{p_1, p_2, \dots, p_M\}$, where $M = |\mathcal{P}|$ is the number of meta-paths considered.

The Heterogeneous Graph Attention Network (HAN) (Wang et al., 2019) is an effective tool for graph representation learning on heterogeneous graphs, capable of assigning different weights to different neighbors and meta-paths via a hierarchical attention mechanism. In this context, we utilize HAN to learn node representations within each meta-path induced subgraph. HAN adaptively aggregates the neighbor information of nodes in each subgraph to integrate both node topology and features. Specifically, for a given node and its associated subgraph, HAN first employs a node-level self-attention mechanism to learn the importance (attention scores) of its neighboring nodes. Then, it adaptively aggregates the feature representations of neighbors based on their attention scores. Finally, a semantic-level attention mechanism is used to combine representations from different meta-paths.

For the k th meta-path based subgraph $\mathcal{G}_{pk}$, the importance $e_{ij}^{G_{pk}}$ of node pair (i, j) can be formulated as follows:

$$e_{ij}^{G_{pk}} = \text{LeakyReLU}(\mathbf{a}_{G_{pk}}^T \cdot [\mathbf{f}_i^{G_{pk}} || \mathbf{f}_j^{G_{pk}}]) \quad (2)$$

where $\mathbf{a}_{G_{pk}}$ is the attention parameter of the k th meta-path. To make the attention coefficients of different meta-paths comparable, we use the softmax function to normalize $e_{ij}^{G_{pk}}$:

$$e_{ij}^{G_{pk}'} = \frac{\exp(e_{ij}^{G_{pk}})}{\sum_{j' \in \mathcal{V}_{hi}^{G_{pk}}} \exp(e_{ij'}^{G_{pk}})} \quad (3)$$

where $\mathcal{V}_{hi}^{G_{pk}}$ represents the set of neighbor nodes of node i in the k th meta-path. Next, we aggregate the feature vectors of all meta-path instances using the ReLU activation function to derive the representation of node i based on the k th meta-path:

$$\mathbf{f}_i^{G_{pk}} = \text{ReLU} \left(\sum_{j \in \mathcal{V}_{hi}^{G_{pk}}} e_{ij}^{G_{pk}'} \cdot \mathbf{f}_j^{G_{pk}} \right) \quad (4)$$

In this study, we introduce a multi-head attention mechanism to enhance the stability of the attention coefficient learning process and mitigate the impact of individual attention heads. Specifically, we independently repeat the attention mechanism K times and concatenate the vector representations learned by each attention head. Thus, the representation $\mathbf{z}_i^{G_{pk}}$ of node i under the k th meta-path is given by:

$$\mathbf{z}_i^{G_{pk}} = ||_{j=1}^K \mathbf{f}_{ij}^{G_{pk}} \quad (5)$$

Generally, each node in the heterogeneous graph contains multiple types of semantic information, and the aggregation of nodes features under a single meta-path can only reflect one aspect of the node's information. To learn more comprehensive node embeddings, it is essential to integrate the semantic information represented by multiple meta-paths. A simple method for aggregating embeddings from different meta-paths is to take the average. However, due to the varying importance of meta-paths to node i in the heterogeneous graph, we adopt an attention mechanism to assign weights to each meta-path pattern and then perform aggregation. This ensures that the embeddings with the most crucial semantic information are given the most significance. More specifically, we learn the importance of each meta-path based subgraph and obtain $|\mathcal{P}|$ sets of node embeddings, where each set represents the weighted aggregation of all nodes under the corresponding meta-path:

$$w_{G_{pk}} = \frac{1}{|\mathcal{V}_h|} \sum_{i \in \mathcal{V}_h} \mathbf{q}^T \cdot \tanh(\mathbf{W}_k \cdot \mathbf{z}_i^{G_{pk}} + \mathbf{b}_k) \quad (6)$$

where $\mathbf{W}_k$, $\mathbf{b}_k$, and $\mathbf{q}_k$ denote the weight matrix, bias vector, and semantic-level attention vector associated with the k th meta-path. $\tanh(\cdot)$ represents the tanh activation function. After obtaining the attention scores of different meta-paths for a specific node, we normalize them using the softmax function, allowing for meaningful comparisons of the semantic information across different meta-paths.

$$w_{G_{pk}'} = \frac{\exp(w_{G_{pk}})}{\sum_{j=1}^{|\mathcal{P}|} \exp(w_{G_{pj}})} \quad (7)$$

Finally, we integrate the node embeddings from different meta-paths based on their respective attention scores. The i th herb node embedding $\mathbf{h}_i$ is as follows:

$$\mathbf{h}_i = \sum_{k=1}^{|\mathcal{P}|} w_{G_{pk}'} \cdot \mathbf{z}_i^{G_{pk}} \quad (8)$$

3.3. 带动态系数的对比学习

上述草药表示可能无法有效捕捉草药之间与毒性相关的相似性和差异性，尤其是在每种草药可能具有多个毒性标签的情况下。因此，我们提出了一种带动态系数的对比学习策略来更新草药表示。传统的基于对比学习的方法旨在最小化同一类别实例之间的距离，同时最大化不同类别实例之间的距离（Chen 等, 2020; Hadsell 等, 2006; Khosla 等, 2020）。然而，在多标签草药毒性预测中，两种草药通常只共享部分毒性标签，并且每种草药具有独特的毒性标签。这种多标签现象比单标签情景复杂得多，使得简单地将两种草药视为“正”或“负”对变得困难。

为了捕捉具有多个毒性标签的草药之间的复杂关系，我们基于毒性标签的语义相似性定义了一个动态系数。与单标签设置中的二值指标不同，该系数提供了一种灵活的方式，能够以数值形式量化草药对之间标签重叠的程度，从而使模型能够考虑多标签毒性预测任务中固有的复杂关系。具体来说，我们衡量两种草药之间的特征相似性如下：

$$\cos(\mathbf{h}_i, \mathbf{h}_j) = \frac{\mathbf{h}_i \cdot \mathbf{h}_j}{\|\mathbf{h}_i\|_2 \|\mathbf{h}_j\|_2} \quad (9)$$

其中 $\cos(\dots, \dots)$ 表示余弦相似度， $\mathbf{h}_i$ 和 $\mathbf{h}_j$ 分别是第 i 个和第 j 个药材的学习到的嵌入， $\|\cdot\|_2$ 表示 L_2 范数。

为了进一步结合有害标签关系，我们将标签语义相似度 β_{ij} 定义为两种草药之间共享有害标签的比例：

$$\beta_{ij} = \frac{\mathbf{y}_i^T \cdot \mathbf{y}_j}{\mathbf{y}_i^T \cdot \mathbf{y}_i} \quad (10)$$

其中 $\mathbf{y}_i \in \{0, 1\}^L$ 和 $\mathbf{y}_j \in \{0, 1\}^L$ 分别是第 i 个和第 j 个药材的多重热毒性标签向量，而 L 是毒性标签的数量。分子 $\mathbf{y}_i^T \cdot \mathbf{y}_j$ 计算两个药材共享毒性标签的数量，而分母 $\mathbf{y}_i^T \cdot \mathbf{y}_i$ 表示第 i 个药材的毒性标签总数。这种基于比例的语义相似度使模型不仅能够区分单标签设置下完全相同或不相交的标签集合，还能够量化多标签场景中特有的部分相似性。

在随后的实验中，我们发现 β 可能不足以作为对比学习的动态系数，因为它无法提供两种药材之间相关性的稳健度量。特别是， β 可能会为毒性标签相似度较低的药材样本分配过大的系数。这个问题可能会降低药材毒性预测的性能。因此，我们引入了一个可调的超参数 t 来调整 β ，以获得更合适的动态系数 δ ：

$$\delta_{ij} = (\beta_{ij})^t \quad (11)$$

具有动态系数 δ_{ij} 的草药对对比学习损失定义如下：

$$\mathcal{L}_{ij}^{con} = -\delta_{ij} \log \frac{\exp(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau)}{\sum_{k=1, k \neq i}^N \exp(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau)} \quad (12)$$

τ 是用于缩放相似性分数的温度超参数， N 是批次中草药的总数。该公式使模型能够强调具有较高毒性相似性的草药对，同时有效地学习具有多个相关毒性标签的草药的表示。这种公式明确利用了草药的多标签毒性，因为每对草药对损失的贡献都按其共享毒性标签的程度进行缩放。

对于给定的一对草药 $(\mathbf{h}_i, \mathbf{h}_j)$ ，较高的语义相似度 β_{ij} 会导致较大的动态系数 δ_{ij} ，进而导致较大的损失 $\mathcal{L}_{ij}^{con}$ 。因此，特征相似度 $\cos(\mathbf{h}_i, \mathbf{h}_j)$ 会被优化为更高的值。相反，如果两个样本没有任何共同标签 ($\beta_{ij} = \delta_{ij} = 0$)，则值 $\mathcal{L}_{ij}^{con}$ 将变为零，其余弦相似度 $\cos(\mathbf{h}_i, \mathbf{h}_j)$ 会被优化为较低的值。带有动态系数的对比学习损失用于 N 草药的计算公式如下：

$$\mathcal{L}^{con} = \sum_{i=1}^N \sum_{j=1, j \neq i}^N \mathcal{L}_{ij}^{con} \quad (13)$$

3.4. 基于加权标签融合的草药毒性预测

对于第 i 种草药，它的表示 $\mathbf{h}_i$ 将通过一个三层 MLP，然后经过 sigmoid 激活函数，以计算所有毒性标签的概率向量：

$$\tilde{\mathbf{y}}_i^{mlp} = \text{sigmoid}(\text{MLP}(\mathbf{h}_i)) \quad (14)$$

其中 $\text{MLP}(\cdot)$ 表示一个三层多层感知器。 $\tilde{\mathbf{y}}_i^{mlp} \in \{0, 1\}^L$ 表示预测概率向量，即第 i 种草药与毒性标签相关的概率。

然而，上述的多层感知机（MLP）可能无法充分利用训练数据中编码的宝贵知识。为了进一步增强毒性预测，我们引入了一种加权标签融合（WLF）策略。对于每种目标草药，我们检索具有相似表示的训练草药。借助对比学习，这些邻近草药更可能与目标草药共享有意义的毒性模式。HerbToxNet以加权方式聚合邻近草药的毒性标签，这不仅补充了 MLP 的输出，还帮助实现更准确的毒性预测。

设 $D = \{(\mathbf{h}_j, \mathbf{y}_j)\}_{j=1}^N$ 表示含有 N 种草药的训练集。给定表示为 $\mathbf{h}_i$ 的第 i 种目标草药，我们通过计算 $\mathbf{h}_i$ 与 D 中所有 $\mathbf{h}_j$ 的余弦相似度，从训练集中检索其相关邻居。相似度超过预定义阈值 σ 的邻居被选中以形成邻居集合：

$$\mathcal{N}_\sigma = \{(\mathbf{h}_j, \mathbf{y}_j) \mid \cos(\mathbf{h}_i, \mathbf{h}_j) \geq \sigma\} \quad (15)$$

我们不是使用 K 近邻，而是基于阈值的策略自适应地选择在表示空间中接近目标草药的邻居。

基于 WLF 的预测针对第 i 种草药是通过聚合所选邻居的标签向量，并根据它们与目标草药的相似性加权得到的：

$$\tilde{\mathbf{y}}_i^{wlf} = \sum_{(\mathbf{h}_j, \mathbf{y}_j) \in \mathcal{N}_\sigma} \mu_j \mathbf{y}_j \quad (16)$$

其中权重 μ_j 定义为：

$$\mu_j = \frac{\exp(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau')}{\sum_{(\mathbf{h}_k, \mathbf{y}_k) \in \mathcal{N}_\sigma} \exp(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau')} \quad (17)$$

τ' 是一个温度超参数，用于控制权重分布的尖锐度。通过这种方式，与第 i 种草药更相似的邻居将被分配更大的权重，这比其他邻居对基于 WLF 的预测有更显著的贡献。

为此，我们将基于 MLP 的预测 $\tilde{\mathbf{y}}_i^{mlp}$ 与基于 WLF 的预测 $\tilde{\mathbf{y}}_i^{wlf}$ 结合如下：

$$\tilde{\mathbf{y}}_i = \alpha \tilde{\mathbf{y}}_i^{wlf} + (1 - \alpha) \tilde{\mathbf{y}}_i^{mlp} \quad (18)$$

其中标量参数 α 平衡了两种预测来源。

我们采用不对称损失（ASL）（Ridnik 等, 2021）作为分类损失，以应对多标签草药毒性预测的类别不平衡问题。ASL 通过对正负样本分配不同的惩罚，有效减轻了标签不平衡的影响，从而增强模型对罕见但关键的毒性标签的关注。该损失

3.3. Contrastive learning with dynamic coefficient

The above herb representations may not effectively capture toxicity-related similarities and differences among herbs, especially under setting where each herb can possess multiple toxic labels. Therefore, we propose a contrastive learning with dynamic coefficient strategy to update the herb representations. Canonical contrastive learning based methods aim to minimize the distance between instances of the same class while maximizing that between instances of different ones (Chen et al., 2020; Hadsell et al., 2006; Khosla et al., 2020). However, in multi-label herb toxicity prediction, two herbs often share only a subset of toxic labels, and each herb posses unique toxic labels. This multi-label phenomenon imposes greater complexity than the single-label setting, making it hard to simply treat two herbs as either “positive” or “negative” pairs.

To capture the intricate relationships between herbs with multiple toxic labels, we define a dynamic coefficient based on toxic label semantic similarity. Unlike binary-valued indicators in single-label settings, this coefficient provides a flexible way to numerically quantify the degree of overlapped labels between herb pairs, allowing the model to account for the complex relationships inherent in multi-label toxicity prediction tasks. Specifically, we measure the feature similarity between two herbs as:

$$\cos(\mathbf{h}_i, \mathbf{h}_j) = \frac{\mathbf{h}_i \cdot \mathbf{h}_j}{\|\mathbf{h}_i\|_2 \|\mathbf{h}_j\|_2} \quad (9)$$

where $\cos(\dots, \dots)$ denotes the cosine similarity, $\mathbf{h}_i$ and $\mathbf{h}_j$ are the learned embeddings of the i th and j th herbs, respectively, and $\|\cdot\|_2$ represents the L_2 norm.

To further incorporate toxic label relationships, we define the label semantic similarity β_{ij} as the fraction of shared toxic labels between two herbs:

$$\beta_{ij} = \frac{\mathbf{y}_i^T \cdot \mathbf{y}_j}{\mathbf{y}_i^T \cdot \mathbf{y}_i} \quad (10)$$

where $\mathbf{y}_i \in \{0, 1\}^L$ and $\mathbf{y}_j \in \{0, 1\}^L$ are the multi-hot toxic label vectors for the i th and j th herbs, respectively, and L is the number of toxic labels. The numerator, $\mathbf{y}_i^T \cdot \mathbf{y}_j$, counts the number of shared toxic labels between the two herbs, while the denominator, $\mathbf{y}_i^T \cdot \mathbf{y}_i$, represents the total number of toxic labels for the i th herb. This ratio-based semantic similarity enables the model to distinguish not only completely identical or disjoint label sets in single-label setting, but also to quantify the partial similarity unique to multi-label scenario.

During subsequent experiments, we found that β may not sufficiently suitable to act as the dynamic coefficient for contrastive learning, as it does not provide a robust measure of relevance between two herbs. In particular, β can assign an excessively large coefficient to herb samples with low toxic label similarity. This issue can degrade the performance of herb toxicity prediction. As a result, we introduce a tunable hyperparameter t to adjust β to obtain a more appropriate dynamic coefficient δ :

$$\delta_{ij} = (\beta_{ij})^t \quad (11)$$

The contrastive learning loss with dynamic coefficient δ_{ij} for a herb pair is defined as:

$$\mathcal{L}_{ij}^{con} = -\delta_{ij} \log \frac{\exp(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau)}{\sum_{k=1, k \neq i}^N \exp(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau)} \quad (12)$$

where τ is the temperature hyperparameter used to scale the similarity scores, and N is the total number of herbs in the batch. This formulation allows the model to emphasize similar herb pairs with higher toxicity similarity while effectively learning representations for herbs with multiple related toxic labels. This formulation explicitly leverages the multi-label toxicity of herbs, since each herb pair’s contribution to the loss is scaled by the extent of their shared toxic labels.

For a given pair of herbs $(\mathbf{h}_i, \mathbf{h}_j)$, a higher semantic similarity β_{ij} results in a larger dynamic coefficient δ_{ij} , which in turn leads to a larger loss $\mathcal{L}_{ij}^{con}$. Consequently, the feature similarity $\cos(\mathbf{h}_i, \mathbf{h}_j)$ is optimized toward a higher value. Conversely, if two samples do not share any labels ($\beta_{ij} = \delta_{ij} = 0$), the value $\mathcal{L}_{ij}^{con}$ becomes zero, and their cosine similarity $\cos(\mathbf{h}_i, \mathbf{h}_j)$ is optimized to a lower value. The contrastive learning loss with dynamic coefficient for N herbs is computed as:

$$\mathcal{L}^{con} = \sum_{i=1}^N \sum_{j=1, j \neq i}^N \mathcal{L}_{ij}^{con} \quad (13)$$

3.4. Herb toxicity prediction based on weighted label fusion

For the i th herb, its representation $\mathbf{h}_i$ is passed through a three-layer MLP, followed by a sigmoid activation function, to compute the probability vector for all toxic labels:

$$\tilde{\mathbf{y}}_i^{mlp} = \text{sigmoid}(\text{MLP}(\mathbf{h}_i)) \quad (14)$$

where $\text{MLP}(\cdot)$ denotes a three-layer MLP. $\tilde{\mathbf{y}}_i^{mlp} \in \{0, 1\}^L$ represents the predicted probability vector that the i th herb is associated with toxic label.

However, the above MLP may not fully utilize the valuable knowledge encoded in the training data. To further enhance the toxicity prediction, we introduce a weighted label fusion (WLF) strategy. For each target herb, we retrieve training herbs with similar representations. By virtue of contrastive learning, these neighbors are more likely to share meaningful toxicity patterns with the target herb. HerbToxNet aggregates the toxic labels of neighborhood herbs in a weighted manner, which complements the MLP’s output and contributes to more accurate toxicity prediction.

Let $D = \{(\mathbf{h}_j, \mathbf{y}_j)\}_{j=1}^N$ denote the training set with N herbs. Given the i th target herb with representation $\mathbf{h}_i$, we retrieve its relevant neighbors from the training set by computing the cosine similarity between $\mathbf{h}_i$ and all $\mathbf{h}_j$ in D . Neighbors whose similarity exceeds a predefined threshold σ are selected to form the neighbor set:

$$\mathcal{N}_\sigma = \{(\mathbf{h}_j, \mathbf{y}_j) \mid \cos(\mathbf{h}_i, \mathbf{h}_j) \geq \sigma\} \quad (15)$$

Instead of using K-nearest neighbors, our threshold-based strategy adaptively selects neighbors that are close to the target herb in the representation space.

The WLF-based prediction for the i th herb is obtained by aggregating the label vectors of the selected neighbors, weighted by their similarities with respect to the target herb:

$$\tilde{\mathbf{y}}_i^{wlf} = \sum_{(\mathbf{h}_j, \mathbf{y}_j) \in \mathcal{N}_\sigma} \mu_j \mathbf{y}_j \quad (16)$$

where the weight μ_j is defined as:

$$\mu_j = \frac{\exp(\cos(\mathbf{h}_i, \mathbf{h}_j)/\tau')}{\sum_{(\mathbf{h}_k, \mathbf{y}_k) \in \mathcal{N}_\sigma} \exp(\cos(\mathbf{h}_i, \mathbf{h}_k)/\tau')} \quad (17)$$

with τ' being a temperature hyperparameter that controls the sharpness of the weight distribution. In this way, neighbors with higher similarity to the i th herb are assigned with larger weights, which more significantly contribute to the WLF-based prediction than other neighbors.

To this end, we combine the MLP-based prediction $\tilde{\mathbf{y}}_i^{mlp}$ with WLF-based prediction $\tilde{\mathbf{y}}_i^{wlf}$ as:

$$\tilde{\mathbf{y}}_i = \alpha \tilde{\mathbf{y}}_i^{wlf} + (1 - \alpha) \tilde{\mathbf{y}}_i^{mlp} \quad (18)$$

where the scalar parameter α balances the two sources of predictions.

We adopt the asymmetric loss (ASL) (Ridnik et al., 2021) as the classification loss to address the imbalanced nature of multi-label herb toxicity prediction. ASL effectively mitigates the impact of label imbalance by assigning different penalties to positive and negative samples, thereby enhancing the model’s focus on rare but critical toxic labels. The loss

函数定义如下：

$$\mathcal{L}^{asl} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^L \left[y_{ij}(1 - \tilde{y}_{ij})^{\gamma^+} \log(\tilde{y}_{ij}) + (1 - y_{ij})(p_{ij}^m)^{\gamma^-} \log(1 - p_{ij}^m) \right] \quad (19)$$

$$p_{ij}^m = \max(\tilde{y}_{ij} - m, 0) \quad (20)$$

其中 γ^+ 和 γ^- 分别是针对正样本和负样本的聚焦参数, $m \geq 0$ 是概率边界。

HerbToxNet 的整体损失函数由 ASL 分类损失和对比学习损失组成, 其公式如下：

$$\mathcal{L} = \mathcal{L}^{asl} + \lambda \mathcal{L}^{con} \quad (21)$$

其中 λ 是用于平衡两种损失的超参数。算法 1 总结了 HerbToxNet 的伪代码。

算法 1 HerbToxNet 的整体流程。输入：草药药理学 $\mathbf{p}$ 、功效 $\mathbf{e}$ ；毒性标签 $\mathbf{y}$ ；元路径集 $\mathcal{P}$ ；节点集 $\mathcal{V}$ ；边集 $\mathcal{E}$ ；模型参数 Θ ；输出：预测的草药毒性标签 $\tilde{\mathbf{y}}$ ；
1: $\triangleright$ 异构图构建 2: 构建节点集 $\mathcal{V}_h$ （草药）、 $\mathcal{V}_c$ （成分）、 $\mathcal{V}_t$ （靶点）及边集 $\mathcal{E}_{hc}, \mathcal{E}_{ht}, \mathcal{E}_{ct}, \mathcal{E}_{hh}$ ； 3: 对每个草药 i ，通过公式 (1) 初始化节点特征 $\mathbf{f}_i$ ； 4: 组装异构图 $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ ； 5: $\triangleright$ 通过 HAN 进行草药表示学习 6: 对每个元路径 $p_k \in \mathcal{P}$ 执行 7: 从 $\mathcal{G}$ 提取元路径子图 $\mathcal{G}_{p_k}$ ； 8: 对每个节点 i 执行 9: 通过公式 (2) 计算注意力分数 $e_{ij}^{G_{p_k}}$ 并通过公式 (3) 进行归一化； 10: 通过公式 (4) 获取邻居聚合特征 $\mathbf{f}_i^{G_{p_k}}$ ； 11: 通过公式 (5) 获得多头特征表示 $\mathbf{z}_i^{G_{p_k}}$ ； 12: 结束 13: 结束 14: 通过公式 (6) 计算元路径注意力 $w_{G_{p_k}}$ ，并通过公式 (7) 归一化； 15: 融合元路径嵌入以得到 $\mathbf{h}_i$ ，公式 (8)； 16: $\triangleright$ 对比学习

(Xu 等, 2021)、HERB (Fang 等, 2021)、SymMap v2 (Wu 等, 2019b)、TCMBank (Lv 等, 2023) 以及 HIT 2.0 (Yan 等, 2022b)。毒性标签来自主要文献和数据库资源, 如 Google Scholar、CNKI、VIP、万方、PubMed 和 Web of Science, 并被分为五类: 心脏毒性、肾毒性、肝毒性、全身毒性及其他毒性。表 1 给出了数据来源的概览。

我们数据集的策展流程在补充文件的第 I 节中进行了说明。然后, 我们将某种草药已知的有毒标签作为该草药的正样本, 其他标签作为负样本。我们将 15% 的草药保留为测试集, 其余草药用于五折交叉验证。该过程独立重复十次以确保稳健性。为了保证完整性, 在分割过程中, 与某种草药相关的所有交互都会被分入同一子集。

4.1.2. 评估指标

为了评估 HerbToxNet 与基线方法的性能, 我们采用了四种广泛使用的评估指标 (Qiu 等, 2023): 宏平均 F1 分数 (Macro-F1)、微平均 F1 分数 (Micro-F1)、平均受试者工作特征曲线下面积 (AvgAUC) 和单一错误 (OneError)。宏平均 F1 分数独立计算每个标签的 F1 分数, 然后取平均值, 反映所有标签的整体表现。微平均 F1 分数汇总所有标签的贡献来计算平均 F1 分数, 反映所有样本的整体表现。AvgAUC 对每个标签计算 AUC, 然后取平均值, 数值越高表示有毒标签与非有毒标签的可区分性越好。OneError 统计预测的排名最高的有毒标签不在真实标签中的次数, 数值越低表示模型越有可能将真实标签排在第一位。为了与其他指标保持一致, 我们使用 1 - OneError 代替该指标。

4.1.3. 基线方法

为了证明我们提出的 HerbToxNet 的有效性, 选择了几种具有代表性的方法作为基线, 这些方法可以总结为三类: 草药毒性预测方法、化学药物毒性预测方法以及生物医学领域的多标签预测方法。(i) 草药毒性预测方法: ZF-LightGBM (Yu 等, 2025) 对 Zhengqing Fengtongning 诱导的肝损伤数据集上的十种机器学习模型进行了比较, 发现 LightGBM 表现最佳; QSAR-ANN 和 QSAR-SVM (Sun 等, 2019) 利用草药肾毒性数据构建 QSAR 模型, 分别应用 ANN 和 SVM 算法进行肾毒性预测; EI-MS-RF (Jia 等, 2024) 通过将草药毒性与电子电离质谱数据的高级分析描述符相关联, 并结合可解释的机器学习模型进行毒性预测。(ii) 化学药物毒性预测方法: DeepDILI (Li 等, 2020) 结合模型级别的

function is defined as follows:

$$\mathcal{L}^{asl} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^L \left[y_{ij}(1 - \tilde{y}_{ij})^{\gamma^+} \log(\tilde{y}_{ij}) + (1 - y_{ij})(p_{ij}^m)^{\gamma^-} \log(1 - p_{ij}^m) \right] \quad (19)$$

$$p_{ij}^m = \max(\tilde{y}_{ij} - m, 0) \quad (20)$$

where γ^+ and γ^- are focusing parameters for positive and negative samples, respectively, and $m \geq 0$ is the probability margin.

The overall loss function of HerbToxNet consists of the ASL classification loss and the contrastive learning loss, formulated as:

$$\mathcal{L} = \mathcal{L}^{asl} + \lambda \mathcal{L}^{con} \quad (21)$$

where λ is a the hyperparameter that balances the two losses. Algorithm 1 summarizes the pseudo code of HerbToxNet

Algorithm 1 The overall process of HerbToxNet.

Input: herb pharmacology $\mathbf{p}$, efficacy $\mathbf{e}$; toxic labels $\mathbf{y}$; meta-path set $\mathcal{P}$; node sets $\mathcal{V}$; edge sets $\mathcal{E}$; model parameters Θ ;

Output: Predicted herb toxic label $\tilde{\mathbf{y}}$;

```

1:  $\triangleright$  Heterogeneous Graph Construction
2:   Build node set  $\mathcal{V}_h$  (herb),  $\mathcal{V}_c$  (ingredient),  $\mathcal{V}_t$  (target) and edge set  $\mathcal{E}_{hc}, \mathcal{E}_{ht}, \mathcal{E}_{ct}, \mathcal{E}_{hh}$ ;
3:   For each herb  $i$ , initialize node features  $\mathbf{f}_i$  via Eq. (1);
4:   Assemble the heterogeneous graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ ;
5:  $\triangleright$  Herb Representation Learning by HAN
6: for each meta-path  $p_k \in \mathcal{P}$  do
7:   Extract meta-path subgraph  $\mathcal{G}_{p_k}$  from  $\mathcal{G}$ ;
8:   for each node  $i$  do
9:     Compute attention score  $e_{ij}^{G_{p_k}}$  via Eq. (2) and normalize it via Eq. (3);
10:    Get neighbor-aggregated features  $\mathbf{f}_i^{G_{p_k}}$  via Eq. (4);
11:    Obtain multi-head feature representation  $\mathbf{z}_i^{G_{p_k}}$  via Eq. (5);
12:  end for
13: end for
14:   Compute meta-path attention  $w_{G_{p_k}}$  via Eq. (6) and normalize it via Eq. (7);
15:   Fuse meta-path embeddings to get  $\mathbf{h}_i$  via Eq. (8);
16:  $\triangleright$  Contrastive Learning with Dynamic Coefficient
17: for each herb pair  $(i, j)$  do
18:   Compute feature similarity via Eq. (9);
19:   Compute semantic similarity  $\beta_{ij}$  and dynamic coefficient  $\delta_{ij}$  via Eq. (10) and Eq. (11);
20: end for
21:   Compute total contrastive loss  $\mathcal{L}^{con}$  via Eq. (13);
22:  $\triangleright$  Herb Toxicity Prediction
23: for each herb  $i$  do
24:   MLP predicts  $\tilde{y}_i^{mlp}$  via Eq. (14);
25:   Weighted fuse  $\tilde{y}_i^{mlp}$  and  $\tilde{y}_i^{wlf}$  into  $\tilde{y}_i$  via Eq. (18);
26: end for
27:   Compute asymmetric loss  $\mathcal{L}^{asl}$  via Eq. (19) and Eq. (20);
28:   Compute total loss  $\mathcal{L}$  via Eq. (21);
29:   Update model parameters  $\Theta$  by back propagation and Adam;
30: return Predicted toxic labels of herb  $\tilde{\mathbf{y}}$ ;
```

4. Experiment

4.1. Experimental setup

4.1.1. Dataset

We collected pharmacological properties, efficacy, and interaction data for herbs from established databases, including ChP (2020)

(Xu et al., 2021), HERB (Fang et al., 2021), SymMap v2 (Wu et al., 2019b), TCMBank (Lv et al., 2023), and HIT 2.0 (Yan et al., 2022b). Toxicity labels were obtained from major literature and database sources such as Google Scholar, CNKI, VIP, WanFang, PubMed, and Web of Science, and were grouped into five categories: cardiotoxicity, nephrotoxicity, hepatotoxicity, systemic toxicity, and other toxicity. Table 1 gives an overview of the data sources.

The curation pipeline of our dataset is reported in Section I of Supplementary file. Then, we take known toxic labels of a herb as positive samples and others as negative ones of this herb. We keep 15% of the herbs as the testing set, while the remaining herbs are used for five-fold cross-validation. This process is independently repeated ten times to ensure robustness. To ensure the integrity, all interactions associated with a herb are grouped into the same subset during the partition process.

4.1.2. Evaluation metrics

To evaluate the performance of HerbToxNet and the baseline methods, we employed four widely used evaluation metrics (Qiu et al., 2023): Macro-averaged F1 Score (Macro-F1), Micro-averaged F1 Score (Micro-F1), Average Area Under the receiver operating characteristic Curve (AvgAUC), and One Error (OneError). Macro-F1 independently calculates the F1 score for each label and then takes the average, reflecting the overall performance across all labels. Micro-F1 aggregates the contributions of all labels to compute the average F1 score, reflecting overall performance on all samples. AvgAUC computes the AUC for each label and then takes the average, a higher value means a better separability between toxic and non-toxic labels. OneError counts the number of the predicted toxic label with top-rank is not among the true labels; a lower values indicate that the model is more likely to rank a true label first. For consistency with other metrics, we use 1 - OneError instead.

4.1.3. Baseline methods

To prove the effectiveness of our proposed HerbToxNet, several representative methods were selected as the baseline, which can be summarized into three categories: herb toxicity prediction methods, chemical drug toxicity prediction methods, and multi-label prediction methods in the field of biomedicine. (i) Herb toxicity prediction methods: ZF-LightGBM (Yu et al., 2025) compares ten machine learning models on Zhengqing Fengtongning-induced liver injury dataset and finds LightGBM performing the best; QSAR-ANN and QSAR-SVM (Sun et al., 2019) uses herb nephrotoxicity data to construct QSAR model, and apply ANN and SVM algorithms to predict nephrotoxicity, respectively; EI-MS-RF (Jia et al., 2024) makes toxicity predictions by correlating herb toxicity with advanced analytical descriptors of electron ionization mass spectrometry data, combined with interpretable machine learning models. (ii) Chemical drug toxicity prediction methods: DeepDILI (Li et al., 2020) combines model-level representations generated by traditional machine learning algorithms with a deep learning framework based on Mold2 descriptors to predict drug-induced liver injury; R-E-GA (Yan et al., 2022a) rotates the fusion vector represented by a high-dimensional molecule in the feature space to find a subset of features with better predictive performance, and then integrates an Adaboost-based ensemble learning method into R-E-GA to improve prediction accuracy; TIRPnet (Wei et al., 2024) is designed to decipher knowledge from abundant of TCM spontaneous reports, enabling the prediction of potential adverse drug reactions (ADRs) for individual ingredient. (iii) multi-label prediction methods: ML-PRDF (Gong et al., 2023) initially utilizes PCC-MLRF (Gong et al., 2023) for feature selection, and then inputs the meticulously selected features into the multi-label deep forest model to conduct differentiation and predictive analysis; MulGRaL (Pham et al., 2022) constructs a heterogeneous information network relevant to the prediction outcomes, subsequently combines multiple multi-label classifiers to enhance the accuracy and reliability of prediction; MLC-CNN (Ge et al., 2021) involves grouping multi-labels according to their inter-label correlations and utilizes neural network models to perform predictions for

表

1

Details of the collected data sources.					
	Type	Size	来源		
Herb	Ph 药物	252×33	第 (2020)、 HERB、 SymMap v2、 TCM- 章 银行, HIT 2.0	logical Pr属性	
	功效	252×331			
	草药-靶点	34988			
	Herb-成分 成分-目标	3275			
		5932			
毒性	心脏毒性	74	谷歌	ic Toxicity	
	肾毒性 肝毒性	101			
		114			
	系统 其他T	167	Scholar CNKI VIP WanFang , , , PubMed WOS		
	oxicity	150			
注意: The size of ‘Efficacy’ is 252×331 indicating that the number of herbs is 252 and the , number of efficacy is 331. ‘Pharmacological Property’ is represented in the same way as “功效”。					

每个组都能有效利用不同特征对不同标签的影响。

由于前两类方法最初是为二分类任务开发的，我们将它们的最终二分类输出修改为生成适合我们任务的多标签预测。此外，由于我们的数据集中没有EI-MS数据，我们用草药的药理特性和功效数据替代EI-MS描述符，并实现了与原始工作中描述的相同的EI-MS-RF流程。对于每个基线方法，我们使用推荐参数作为起点，并根据我们的数据集进行必要的微调。所有基线方法和我们的HerbToxNet的详细参数，以及HerbToxNet的全面参数敏感性分析，均在补充文件的第三节中进行了介绍。

4.2. 与现有解决方案的结果比较

表 2 总结了 HerbToxNet 和基线方法的性能。HerbToxNet 在大多数数评估指标上实现了最佳的草药毒性预测性能。其他重要观察包括：

(i) HerbToxNet 在草药毒性预测方面显著优于其他方法，包括 ZF-LightGBM、QSAR-SVM、EI-MS-RF 和 QSAR-ANN，主要原因是这些方法无法有效捕捉草药的复杂多成分和多靶点特性。该缺陷限制了它们的泛化能力，在处理多标签毒性预测任务时尤为明显。此外，它们通常只预测单一毒性标签，无法有效建模毒性标签之间的复杂关联，从而限制了其在多标签毒性预测中的表现。相比之下，HerbToxNet 在复杂的多标签毒性预测任务中表现更佳，在大多数评估指标上显著领先。其优势如下。首先，HerbToxNet 构建了一个草药-成分-靶点异质网络，并利用元路径提取属性和关系

草药的功能信息，实现了对草药复杂特征的精确建模。其次，它引入了一种带动态系数策略的对比学习方法，使毒性标签相近的草药对表示更为接近，同时将毒性标签不同的草药对分开，从而增强模型对多标签任务的适应能力。第三，HerbToxNet 引入了一种加权标签融合策略，有效利用相似草药的毒性标签，从而进一步提高预测性能，并缓解草药数据稀缺带来的影响。(ii) HerbToxNet 在预测药物毒性的方法上也显著优于包括 DeepDILI、R-E-GA 和 TIRPnet 在内的方法。这些方法在草药多标签毒性预测任务中表现不佳，因为化学药物通常具有明确定义的分子结构和指纹，这为预测提供了丰富而详细的信息。相比之下，草药由多种成分组成

(iii) 基于多标签分类的方法在预测草药的毒性标签方面相比其他两种方法显示出一定的优势。这是因为它们可以同时考虑多个毒性标签，这自然适用于多标签草药毒性预测。尽管这些方法在某些场景下能够提供有竞争力的结果，但在挖掘有限草药数据的内在信息和复杂特征方面可能存在局限。因此，如表2所示，这些方法再次不如HerbToxNet。HerbToxNet性能优越的主要原因在于其有效的建模。

2		城			
Performance comparison of HerbToxNet and baseline methods in predicting herb toxicity.					
	Macro F1	微型F1	AvgAUC	1 - 一个错误	
ZF-李ghtGBM	0.237 ±0.04 13	0.4127±0.04 20	0.5163±0.0307	0.2535±0.0487	
QSAR-ANN	0.4195±0.0460	0.4510±0.0429	0.5105±0.0343	0.1936±0.0536	
QSAR-SV M	0.1619±0.0235	0.3842±0.04 22	0.5132±0.04 22	0.2582±0.0520	
EI-MS-RF	0.3432±0.0384	0.4437±0.0304	0.5120±0.0370	0.2242±0.048 2	
DeepDILI	0.3035±0.0956	0.4756±0.0842	0.5166±0.0489	0.5329±0.070 2	
R-E-GA	0.2894±0.0470	0.4220±0.0465	0.5092±0.0352	0.2313±0.0417	
TIRP网络	0.4089±0.0420	0.4265±0.0468	0.5058±0.0384	0.2111±0.0608	
ML-PRDF	0.4059±0.0327	0.4508±0.0439	0.5294±0.0548	0.4711±0.0698	
MulGRaL	0.4135±0.0421	0.5576±0.0325	0.5606±0.0209	0.5195±0.0611	
MLC-CNN	0.4966±0.0653	0.5086±0.0508	0.5525±0.0349	0.5115±0.0676	
	0.5652± 0.0161	0.6247±0.0172	0.6003± 0.0141	0.5587±0.0485	
注意:					
The best results are highlighted in bold font with statistical significance checked , % by student <i>t</i> -test at 95 level.					

Table 1
Details of the collected data sources.

	Type	Size	Sources
Herb	Pharmacological Property	252×33	Chp (2020), HERB, SymMap v2, TCM-Bank, HIT 2.0
	Efficacy	252×331	
	Herb-Target	34988	
	Herb-Ingredient	3275	
		5932	
Toxicity	Cardiotoxicity	74	Google Scholar, CNKI, VIP, WanFang, PubMed, WOS
	Nephrotoxicity	101	
	Hepatotoxicity	114	
	Systemic Toxicity	167	
	Other Toxicity	150	

Note: The size of ‘Efficacy’ is 252×331, indicating that the number of herbs is 252 and the number of efficacy is 331. ‘Pharmacological Property’ is represented in the same way as ‘Efficacy’.

each group, effectively leveraging the impact of diverse features on distinct labels.

Since the first two categories of methods were originally developed for binary classification tasks, we modify their final binary outputs to produce multi-label predictions suitable for our task. In addition, as EI-MS data are not available in our dataset, we substitute the EI-MS descriptors with herb pharmacological property and efficacy data, and implement the same EI-MS-RF pipeline as described in the original work. For each baseline method, we used the recommended parameters as a starting point and fine-tuned them as necessary for our dataset. Detailed parameters for all baselines and our HerbToxNet, as well as a comprehensive parameter sensitivity analysis of HerbToxNet, are presented in Section III of Supplementary file.

4.2. Results comparison with existing solutions

Table 2 summarizes the performance of HerbToxNet and baselines. HerbToxNet achieves the best herb toxicity prediction performance in most evaluation metrics. Other important observations are: (i) HerbToxNet significantly outperforms other herb toxicity prediction methods, including ZF-LightGBM, QSAR-SVM, EI-MS-RF, and QSAR-ANN, primarily due to they inability to effectively capture the complex multi-ingredient and multi-target characteristics of herbs. This defect limits their generalization ability, which is particularly evident when handling multi-label toxicity prediction tasks. Moreover, they typically predict single toxic labels and fail to effectively model the intricate correlations between toxic labels, thereby restricting their performance in multi-label toxicity prediction. In contrast, HerbToxNet achieves better performance in complex multi-label toxicity prediction tasks, significantly leading in most evaluation metrics. Its advantages are as follows. First, HerbToxNet constructs a herb-ingredient-target heterogeneous network and utilizes meta-path to extract properties and rela-

Table 2
Performance comparison of HerbToxNet and baseline methods in predicting herb toxicity.

	Macro F1	Micro F1	AvgAUC	1 - OneError
ZF-LightGBM	0.2371±0.0413	0.4127±0.0420	0.5163±0.0307	0.2535±0.0487
QSAR-ANN	0.4195±0.0460	0.4510±0.0429	0.5105±0.0343	0.1936±0.0536
QSAR-SVM	0.1619±0.0235	0.3842±0.0422	0.5132±0.0422	0.2582±0.0520
EI-MS-RF	0.3432±0.0384	0.4437±0.0304	0.5120±0.0370	0.2242±0.0482
DeepDILI	0.3035±0.0956	0.4756±0.0842	0.5166±0.0489	0.5329±0.0702
R-E-GA	0.2894±0.0470	0.4220±0.0465	0.5092±0.0352	0.2313±0.0417
TIRPnet	0.4089±0.0420	0.4265±0.0468	0.5058±0.0384	0.2111±0.0608
ML-PRDF	0.4059±0.0327	0.4508±0.0439	0.5294±0.0548	0.4711±0.0698
MulGRaL	0.4135±0.0421	0.5576±0.0325	0.5606±0.0209	0.5195±0.0611
MLC-CNN	0.4966±0.0653	0.5086±0.0508	0.5525±0.0349	0.5115±0.0676
	0.5652± 0.0161	0.6247±0.0172	0.6003± 0.0141	0.5587±0.0485

Note: The best results are highlighted in bold font, with statistical significance checked by student *t*-test at 95 % level.

表 3

Comparison of HerbToxNet with other methods on hepatotoxicity and nephrotoxicity prediction.						
	H肝毒性 Pr预言			Nephro毒性 Pr预言		
	AUROC	AUPRC	F1	AUROC	AUPRC	F1
ZF-李ghtGBM	0.5610±0.0803	0.5369±0.0944	0.6323±0.0565	0.5682±0.0554	0.4668±0.0723	0.5881±0.0577
QSAR-ANN	0.5543±0.0825	0.5409±0.1038	0.6427±0.0565	0.5571±0.0738	0.4847±0.0798	0.5973±0.0609
QSAR-SV M	0.5675±0.1038	0.5753±0.1197	0.6295±0.0574	0.5695±0.0649	0.4913±0.0695	0.5848±0.0585
EI-MS-RF	0.5624±0.0765	0.5304±0.0984	0.0.0594	0.5431±0.0743	0.4529±0.0782	0.6023±0.0566
DeepDILI	0.5486±0.0763	0.5658±0.1005	0.2720.0580	0.5799±0.0744	0.4989±0.0782	0.5949±0.0550
R-E-GA	0.5342±0.0786	0.5437±0.1011	0.6218±0.0574	0.5627±0.0804	0.5053±0.0939	0.5728±0.0578
TIRP网络	0.5269±0.0824	0.4959±0.0907	0.5084±0.0918	0.4949±0.0722	0.4279±0.0735	0.4630±0.0761
草本毒素Net	0.7060±0.0613	0.7698±0.0718	0.7356±0.0170	0.7741±0.0268	0.6381±0.0419	0.6520±0.0293

注意:The best results are highlighted in bold font, with statistical significance checked by student *t*-test at 95 %level.

通过异质网络、对比学习和加权标签融合对草药、成分和靶点进行研究。

4.3 肝毒性和肾毒性预测

HerbToxNet 旨在进行多标签毒性预测，而大多数方法针对的是草药 (Jia 等, 2024; Sun 等, 2019; Yu 等, 2025) 和药物 (Li 等, 2020; Wei 等, 2024; Yan 等, 2022a) 的单标签毒性预测，它们建立在单类型毒性的二分类任务上。为了评估 HerbToxNet 是否可以适应这些二分类任务，我们从数据集中分离出肝毒性和肾毒性草药，将它们分别划分为肝毒性和肾毒性数据集，并在两个数据集中评估我们的方法。其中，肝毒性数据集包含 114 种草药，肾毒性数据集包含 101 种草药。由于这是一个二分类任务，因此采用 AUROC、AUPRC 和 F1 进行评估。

如表3所示，尽管HerbToxNet最初是为多标签毒性预测设计的，但在应用于针对单一毒性类型的二分类任务时，其表现出良好的适应性和稳健性。在肝毒性预测任务中，HerbToxNet在所有评估指标上均取得最佳表现，优于所有基线模型。与现有方法如Deep-DILI、基于QSAR的方法以及EI-MS-RF相比，HerbToxNet获得了更高的AUPRC值，表明其在类别不平衡的情况下识别阳性样本的能力更强。同样，在肾毒性预测任务中，HerbToxNet也保持了竞争力，取得了最高的AUROC、AUPRC和F1分数，这表明其在识别肾毒性草药方面具有很强的区分能力。值得注意的是，HerbToxNet在两种毒性类型上均获得了持续更好的结果，验证了其在不同毒理任务中的稳定适应性和学习能力。

这些结果强调了 HerbToxNet 不仅在建模复杂的多标签毒性关系方面有效，而且能够提取关键特征并在二元分类任务中评估毒性风险。

4.4. 药物毒性预测

HerbToxNet 不仅限于草药毒性预测，它还可以扩展到化学药物。为了研究其泛化能力，我们在 UniTox 数据集 (Silberg 等, 2024) 上进行了评估，该数据集包含 1349 种化学化合物，并标注了八种毒性标签。鉴于化学药物与草药在结构和语义上的差异，我们移除了异质图和 HAN，直接使用每种药物的 SMILES 表示作为输入。从 SMILES 字符串中，我们提取 Morgan 指纹来表示分子特征。然后，这些特征与对比学习模块和加权标签融合策略相结合，以对化学药物进行毒性预测。在基线方法中，MulFRaL 被排除在比较之外，因为它高度依赖于异质图结构，而在这该设置中不可行。其他所有基线方法保持不变。实验设置和评估指标与用于草药的相同。

表4总结了HerbToxNet及基线方法在药物毒性预测上的表现。在大多数评估指标上，HerbToxNet的结果优于基线方法。它在化学药物毒性预测任务中再次展现了卓越的性能，这主要归功于其在特征表示和标签预测方面的方法优势。尽管在此设置下排除了异构图和HAN，HerbToxNet仍利用对比学习模块来增强药物表示的区分能力。具体而言，对比学习策略促使具有相似毒性特征的药物在特征空间中更接近，这有助于区分不同的毒性标签。Morgan指纹的使用提供了一种简洁且适用范围广的分子表示，这使得模型即使在缺乏特定于中药复杂语义上下文的情况下，也能保持较高的预测效能。此外，

of herbs, ingredients and targets by heterogeneous network, contrastive learning and weighted label fusion.

4.3. Hepatotoxicity and nephrotoxicity prediction

HerbToxNet is designed for multi-label toxicity prediction, while most methods target at single-label toxicity prediction of herbs (Jia et al., 2024; Sun et al., 2019; Yu et al., 2025) and of drugs (Li et al., 2020; Wei et al., 2024; Yan et al., 2022a), they build on binary tasks for single type toxicity. To assess whether HerbToxNet can be adapted to these binary tasks, we isolated hepatotoxicity and nephrotoxicity herbs from the dataset, divided them as hepatotoxicity and nephrotoxicity datasets and evaluated our approach in both datasets. Among them, the hepatotoxicity dataset contains 114 herbs and the nephrotoxicity dataset contains 101 herbs. Since this is a binary task, AUROC, AUPRC, and F1 are used for evaluation.

As shown in Table 3, although HerbToxNet is originally designed for multi-label toxicity prediction, it demonstrates good adaptability and robustness when applied to binary classification tasks focused on individual toxicity types. In the hepatotoxicity prediction task, HerbToxNet achieves the best performance in all evaluation metrics, outperforming all baseline models. Compared with existing methods such as Deep-DILI, QSAR-based approaches, and EI-MS-RF, HerbToxNet achieves higher AUPRC values, suggesting a better capability to identify positive samples under class-imbalanced settings. Similarly, HerbToxNet holds competitive performance in the nephrotoxicity prediction task, achieving the highest AUROC, AUPRC, and F1 scores, which demonstrates its strong discriminative ability in identifying nephrotoxic herbs. Notably, HerbToxNet obtains consistently better results across both toxicity types, validating its stable adaptability and learning capacity across different toxicological endpoints.

These results underscore that HerbToxNet is not only effective in modeling complex multi-label toxicity relationships, but also capable of extracting critical features and assessing toxicity risk in binary classification tasks.

Table 3
Comparison of HerbToxNet with other methods on hepatotoxicity and nephrotoxicity prediction.

	Hepatotoxicity Prediction			Nephrotoxicity Prediction		
	AUROC	AUPRC	F1	AUROC	AUPRC	F1
ZF-LightGBM	0.5610±0.0803	0.5369±0.0944	0.6323±0.0565	0.5682±0.0554	0.4668±0.0723	0.5881±0.0577
QSAR-ANN	0.5543±0.0825	0.5409±0.1038	0.6427±0.0565	0.5571±0.0738	0.4847±0.0798	0.5973±0.0609
QSAR-SVM	0.5675±0.1038	0.5753±0.1197	0.6295±0.0574	0.5695±0.0649	0.4913±0.0695	0.5848±0.0585
EI-MS-RF	0.5624±0.0765	0.5304±0.0984	0.6435±0.0594	0.5431±0.0743	0.4520±0.0782	0.6023±0.0566
DeepDILI	0.5486±0.0763	0.5658±0.1005	0.6272±0.0580	0.5799±0.0744	0.4989±0.0782	0.5949±0.0550
R-E-GA	0.5342±0.0786	0.5437±0.1011	0.6218±0.0574	0.5627±0.0804	0.5053±0.0939	0.5728±0.0578
TIRPnet	0.5269±0.0824	0.4959±0.0907	0.5084±0.0918	0.4949±0.0722	0.4279±0.0735	0.4630±0.0761
HerbToxNet	0.7060±0.0613	0.7698±0.0718	0.7356±0.0170	0.7741±0.0268	0.6381±0.0419	0.6520±0.0293

Note: The best results are highlighted in bold font, with statistical significance checked by student *t*-test at 95 % level.

4.4. Drug toxicity prediction

HerbToxNet is not limited to herb toxicity prediction, it can also be extended to chemical drugs. To study its generalizability, we evaluate it on the UniTox dataset (Silberg et al., 2024), which contains 1349 chemical compounds annotated with eight toxic labels. Given the structural and semantic differences between chemical drugs and herbs, we remove the heterogeneous graph and HAN, and directly use the SMILES representation of each drug as input. From the SMILES strings, we extract Morgan fingerprints to represent molecular features. These features are then integrated with the contrastive learning module and the weighted label fusion strategy to perform toxicity prediction for chemical drugs. Among the baseline methods, MulFRaL was excluded from comparison as it relies heavily on heterogeneous graph structures, which are not applicable in this setting. All other baseline methods keep unchanged. The experimental setup and evaluation metrics are the same as those used in the herb toxicity prediction task.

Table 4 summarizes the performance of HerbToxNet and baselines on predicting drug toxicity. The results of HerbToxNet are better than those of baseline methods in most evaluation metrics. It again achieves superior performance in the toxicity prediction task for chemical drugs, primarily due to its methodological strengths in both feature representation and label prediction. Although the heterogeneous graph and HAN are excluded in this setting, HerbToxNet leverages the contrastive learning module to enhance the discriminability of drug representations. Specifically, the contrastive learning strategy encourages representations of drugs with similar toxicity profiles to be closer in the feature space, this helps to distinguish different toxic labels. The use of Morgan fingerprints offers a compact and widely applicable molecular representation, which enables the model to maintain high prediction efficacy even in the absence of complex semantic contexts specific to TCM. Moreover, WLF combines the predictions from MLP and those aggregated from similar herbs, effectively alleviating the issue of limited training data. These results not only validate the generalizability of HerbToxNet, but

表格 4
Performance comparison of HerbToxNet and baseline methods in predicting drug toxicity.

	Macro F1	微型F1	AvgAUC	1 - 一个错误
ZF-李ghtGBM	0.6107±0.0169	0.6961±0.0114	0.6973±0.0139	0.3610±0.0226
QSAR-ANN	0.6196±0.0155	0.7160±0.0128	0.7059±0.0163	0.3250±0.0223
QSAR-SV M	0.5241±0.0212	0.7085±0.0109	0.7038±0.0172	0.3424±0.0281
EI-MS-RF	0.5696±0.0155	0.7129±0.0110	0.7038±0.0148	0.3689±0.0245
DeepDILI	0.5929±0.0174	0.7184±0.0132	0.7055±0.0173	0.8638±0.0234
R-E-GA	0.5382±0.0360	0.6538±0.0303	0.5998±0.0055	0.3714±0.0186
TIRPnet	0.5861±0.0383	0.6905±0.0162	0.6618±0.0226	0.3658±0.0563
ML-PRDF	0.6107±0.0159	0.7080±0.0117	0.6260±0.0129	0.8504±0.0149
MLC-CNN	0.5091±0.0682	0.6648±0.0301	0.6179±0.0265	0.8718±0.0186
草本毒素Net	0.6365±0.0207	0.7251±0.0134	0.7177±0.0141	0.8786±0.0164

注意:

The best results are highlighted in bold font with statistical significance checked , % by student *t*-test at 95 % level.

Table 4
Performance comparison of HerbToxNet and baseline methods in predicting drug toxicity.

	Macro F1	Micro F1	AvgAUC	1 - OneError
ZF-LightGBM	0.6107±0.0169	0.6961±0.0114	0.6973±0.0139	0.3610±0.0226
QSAR-ANN	0.6196±0.0155	0.7160±0.0128	0.7059±0.0163	0.3250±0.0223
QSAR-SVM	0.5241±0.0212	0.7085±0.0109	0.7038±0.0172	0.3424±0.0281
EI-MS-RF	0.5696±0.0155	0.7129±0.0110	0.7038±0.0148	0.3689±0.0245
DeepDILI	0.5929±0.0174	0.7184±0.0132	0.7055±0.0173	0.8638±0.0234
R-E-GA	0.5382±0.0360	0.6538±0.0303	0.5998±0.0055	0.3714±0.0186
TIRPnet	0.5861±0.0383	0.6905±0.0162	0.6618±0.0226	0.3658±0.0563
ML-PRDF	0.6107±0.0159	0.7080±0.0117	0.6260±0.0129	0.8504±0.0149
MLC-CNN	0.5091±0.0682	0.6648±0.0301	0.6179±0.0265	0.8718±0.0186
HerbToxNet	0.6365±0.0207	0.7251±0.0134	0.7177±0.0141	0.8786±0.0164

Note: The best results are highlighted in bold font, with statistical significance checked by student *t*-test at 95 % level.

还强调了其核心设计在不同毒性预测情景下的方法适应性。

4.5. 消融实验与参数敏感性分析

为了进一步研究 HerbToxNet 的贡献因素，我们引入了五个变体：(i) w/o HAN 直接使用草药的属性和功效数据作为特征，而不构建异构图和元路径，但结合对比学习和加权标签融合策略进行预测。(ii) w/o PE 在异图中随机初始化草药节点特征，而不是使用药理属性和功效数据作为初始特征。(iii) w/o CL 在训练阶段移除带有动态系数的对比学习模块。(iv) w/o δ 用毒性标签相似性 β_{ij} 替换动态系数 δ_{ij} 。(v) w/o WLF 仅使用 MLP 进行预测，不采用加权标签融合策略。详细内容见补充文件的第二部分。总体而言，本方法明显优于其变体，强调了在 HerbToxNet 中每个移除因素对于草药毒性预测的重要作用。

我们还进行了不同的实验以研究 HerbToxNet 的参数敏感性。结果在附录文件的第三部分进行了报告和分析。

4.6. 草药毒性预测

在本案例研究中，我们利用 HerbToxNet 预测和分析了经典草药的毒性，包括柴胡、何首乌、巴豆、附子和蓖麻子，在真实世界场景中的表现。为了严格评估模型的泛化能力，数据集中所有其他草药被用作训练集，而选定的五种草药作为独立的测试集进行毒性预测。预测的毒性与已知的毒性标签进行了比较，并从 PubMed 检索了相关文献，以进一步验证模型预测结果与当前研究发现及传统中医理论的一致性。

结果表明，HerbToxNet 能够准确识别典型肝毒性草药（如柴胡和何首乌）的关键毒性，有效揭示极“寒”或“热”草药（如巴豆和附子）的毒性风险，并发现传统上归类为平性（“平”）草药（如蓖麻子）的潜在毒性。此外，HerbToxNet 生成的毒性预测在很大程度上与既有的毒理学证据及经典中医分类体系一致，这表明该模型能够利用传统草药特性（性味、归经）中编码的信息来评估毒性风险。

实验程序的详细描述、病例预测结果以及综合分析在补充材料第四节中提供。

4.7. 可解释性分析

4.7.1. 模型可解释性分析

为了捕捉草药、成分和靶点之间的复杂关系，我们采用了异构注意力网络（HAN）（Wang 等，2019），其特别突出的特点是分层注意力机制。该机制使模型能够在学习节点嵌入时，同时捕捉节点邻居的重要性和不同元路径的相关性。为了更深入地了解模型的可解释性，并更好地理解邻居和元路径的相对重要性，我们对分层注意力机制进行了详细分析。具体而言，我们以柴胡为案例研究，并可视化节点与其邻居在不同元路径下的注意力得分，如图 2 所示。从语义层次的分层注意力角度来看，HAN 对三条元路径（HH、HH 和 HTH）赋予了不同的注意力权重，反映了模型在表示时对不同语义关系的关注。

also highlight the methodological adaptability of its core design across different toxicity prediction scenarios.

4.5. Ablation study and parameter sensitivity analysis

To further study the contribution factors of HerbToxNet, we introduce five variants: (i) w/o HAN directly uses the properties and efficacy data of herb as features without constructing heterogeneous graph and the meta-path, but combined with contrastive learning and weighted label fusion strategies for prediction. (ii) w/o PE randomly initializes the herb node features in the heterogeneous graph instead of using pharmacological property and efficacy data as initial features. (iii) w/o CL removes the contrastive learning module with dynamic coefficients in the training phase. (iv) w/o δ replaces the dynamic coefficient δ_{ij} with the toxicity label similarity β_{ij} . (v) w/o WLF only uses the MLP for prediction, without incorporating the weighted label fusion strategy. The details are reported in Section II of Supplementary file. Overall, outperforms its variants by a clear margin, underscoring the vital role of each removed factor in HerbToxNet for the prediction of herb toxicity.

We also conduct different experiments to study the parameter sensitivity of HerbToxNet. The results are reported and analyzed in Section III of Supplementary file.

4.6. Toxicity prediction of herbs

In this case study, we utilized HerbToxNet to predict and analyze the toxicity of canonical herbs, including *Chai Hu*, *He Shou Wu*, *Ba Dou*, *Fu Zi*, and *Bi Ma Zi*, in real-world scenarios. To rigorously evaluate the model's generalization capability, all other herbs in the dataset were used as the training set, while the five selected herbs were treated as a separate test set for toxicity prediction. The predicted toxicities were compared with known toxic labels, and relevant literature was retrieved from PubMed to further verify the consistency of model predictions with current research findings and traditional TCM theory.

The results demonstrate that HerbToxNet can accurately identify key toxicities of canonical hepatotoxic herbs (such as *Chai Hu* and *He Shou Wu*), effectively reveal toxicity risks in extremely “cold” or “hot” herbs (such as *Ba Dou* and *Fu Zi*), and uncover potential toxicity in herbs traditionally classified as neutral (“ping”) (such as *Bi Ma Zi*). Moreover, the toxicity predictions generated by HerbToxNet are broadly consistent with both established toxicological evidence and the classical TCM classification system, indicating that the model can leverage information encoded in traditional herb properties (nature, flavor, meridian tropism) to assess toxicity risk.

The detailed description of the experimental procedures, case prediction results, and comprehensive analyses are provided in Supplementary Section IV.

4.7. Interpretability analysis

4.7.1. Model interpretability analysis

To capture the intricate relationships between herbs, ingredients and targets, we adopt the Heterogeneous Attention Network (HAN) (Wang et al., 2019), which is particularly notable for its hierarchical attention mechanism. This mechanism enables the model to capture both the importance of a node's neighbors and the relevance of different meta-paths when learning node embeddings. To gain deeper insight into the model interpretability and to better understand the relative significance of neighbors and meta-paths, we conduct a detailed analysis of the hierarchical attention mechanism. Specifically, we take *Chai Hu* as a case study and visualize the attention scores between the node and its neighbors under different meta-paths, as illustrated in Fig. 2. From the perspective of semantic-level hierarchical attention, HAN assigns distinct attention weights to three meta-paths (HH, HH and HTH), reflecting the model's emphasis on different semantic relationships when representing the herb node *Chai Hu*. Among them, the HH meta-path receives the highest semantic attention score, indicating that herbs with shared ingredients play a dominant role in the feature learning of *Chai Hu*. In contrast, the HH meta-path receives a moderate weight, while the HTH meta-path has the lowest score. This distribution of attention scores aligns with the connectivity patterns in our dataset. The HH meta-path, with fewer but more specific associations, enables the model to capture distinctive ingredient-level features. On the other hand, the HTH meta-path, involving far more connections, leads to denser but more redundant interactions, diminishing its utility for precise herb representation. The intermediate weight for HH reflects that these herb-herb associations are indirectly inferred from pharmacological and efficacy similarities, making them less specific than HH but less redundant than HTH.

In the node-level attention mechanism, we select the top three nodes with the highest attention scores under each meta-path. For HH meta-path, *Chai Hu* is connected to *San Bai Cao*, itself, and *Feng Du*. All three herbs share the characteristic of a pungent flavor. Regarding meridian tropism, both *Chai Hu* and *Feng Du* target the liver meridian. Although *San Bai Cao* primarily targets the lung meridian, the lung and liver are physiologically interrelated, allowing for complementary therapeutic effects. Functionally, all three exhibit anti-inflammatory, anti-swelling, and analgesic properties. In addition, the connection to the self-node

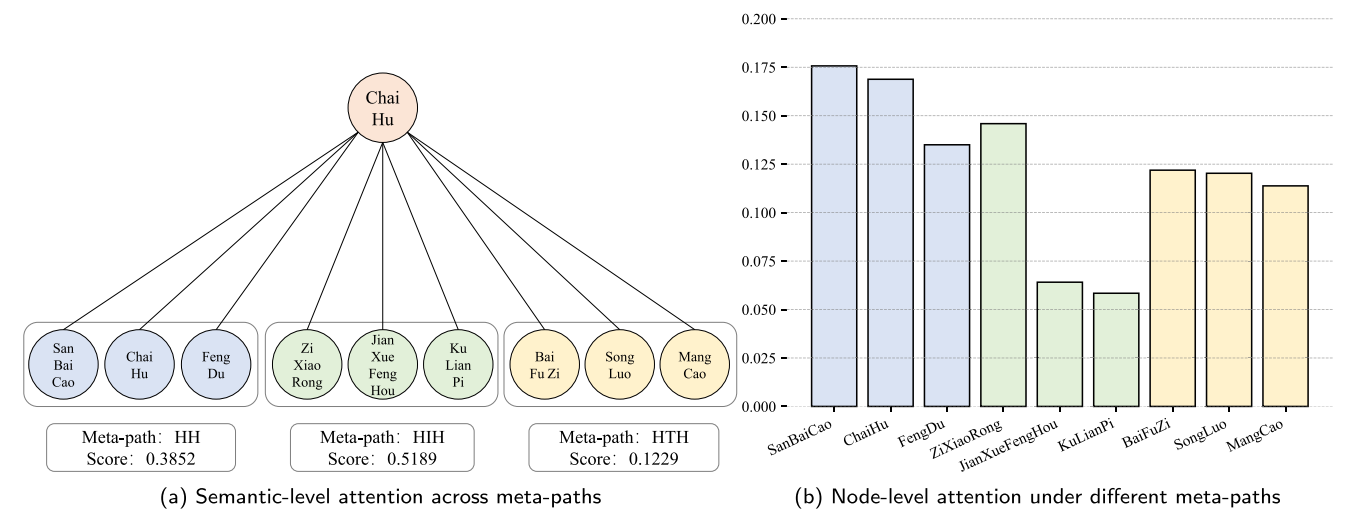
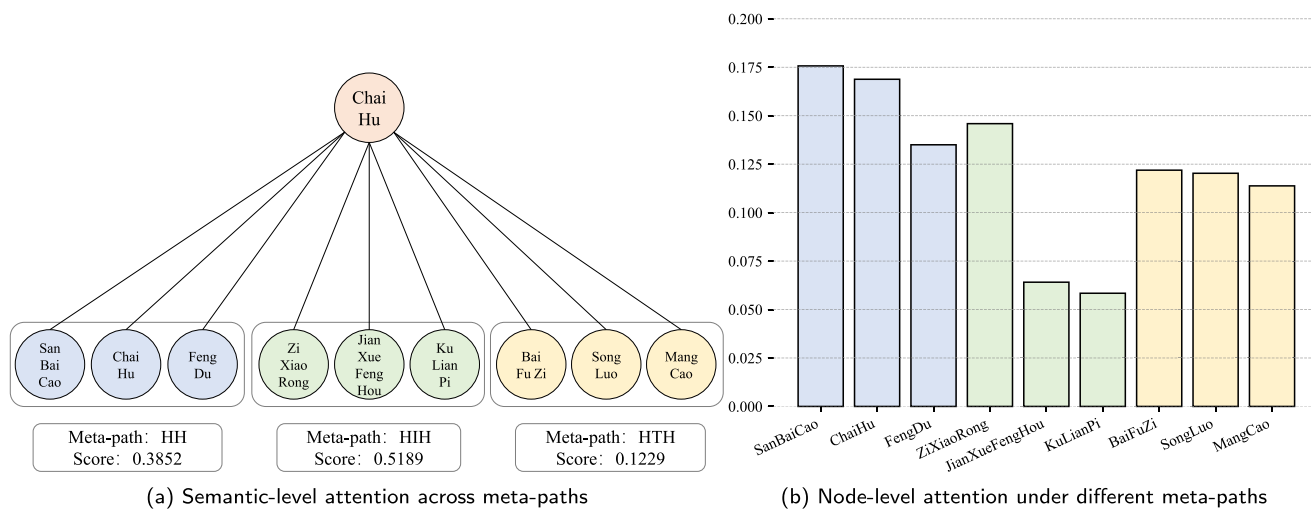


图 2. Hierarchical attention analysis results of HerbToxNet on the herb node *Chai Hu* including (a) semantic-level attention across meta-paths and (b) node-level attention under different meta-paths.

进一步表明，节点本身在学习过程中生成自身表示时起着关键作用。

对于 HIH 元路径，柴胡与紫小蓉、建雪蜂侯和苦楝皮相连。柴胡、紫小蓉、苦楝皮和建雪蜂侯具有相同类别的次生代谢产物，包括三萜皂苷、黄酮类、酚类化合物、挥发油和苦味生物碱。例如，柴胡含有柴胡皂苷，紫小蓉富含黄酮类和酚酸类物质，苦楝皮含有苦楝素及相关的类橘皮素，而建雪蜂侯含有心脏毒性心甘如苦参碱。尽管它们的结构各异，这些成分在生物活性方面具有重叠，尤其体现在抗炎、抗肿瘤和解毒作用上。

对于 HTH 元路径，柴胡与白附子、松萝和蟒草相连。这些草药与柴胡共享几个潜在的药理靶点，包括 TNF、IL6、PTGS2 和 MAPK1，这些靶点与肝损伤和炎症反应密切相关。柴胡中的柴胡皂苷和白附子中的二萜生物碱可能通过激活 NFKB 信号通路上调促炎性细胞因子如 TNF 和 IL6。松萝中的乌司酸已被认为会导致线粒体功能障碍和氧化应激，从而通过 PTGS2 过表达引发肝毒性。蟒草中的生物碱可能影响 MAPK1 信号通路，该通路参与肝细胞凋亡和应激反应。

总之，节点级注意力机制根据共享的草药属性、成分和靶点突出关键关联，为柴胡的模型预测提供了生物学上有意义且可解释的见解。

4.7.2. 生物可解释性分析

为了增强 HerbToxNet 的生物可解释性，我们沿着“草药-成分-靶点-生物通路-毒性”轴构建了一个多层次解释框架。以柴胡为代表案例，我们通过整合元路径分析、文献整理和毒性数据库，系统识别了其有毒成分、相关分子靶点及相关毒性类型。我们进一步通过其生物通路和功能注释将这些组成部分联系起来。我们的结果显示，HerbToxNet 成功揭示了涉及肝毒性、肾毒性和系统毒性的关键有毒成分及其分子靶点，并揭示了与草药诱导毒性反应相关的生物通路和机制。这表明，HerbToxNet 可以为其毒性预测提供机制上合理且生物学上可行的解释，从而支持对草药进行更可靠的风险评估。全面的

5. 结论

在本文中，我们构建了一个多标签草药毒性数据集，并提出了 HerbToxNet，这是一种整合异质网络建模、对比学习和加权标签融合的草药毒性预测框架。实证研究表明，HerbToxNet 的性能优于其他竞争方法。HerbToxNet 可以识别潜在的新毒性并改善草药的临床安全使用。此外，它还能发现与毒性相关的成分和靶点，提供毒性机制的见解，指导解毒剂开发，支持毒性相关数据库的建设以及更安全草药的发现。尽管 HerbToxNet 的表现优于对比方法并提供了一些真实结果，但它仍存在局限性。我们的数据集目前涵盖五类毒性，并且许多草药被分配了多个标签，从而支持多标签分类。由于标准化数据有限，更广泛的毒性类别尚未涵盖。

CRedit 作者贡献声明

朱雍正：数据整理、正式分析、调查、软件、验证、原始稿撰写；缪云博：正式分析、软件、验证、原始稿撰写；孙荣：监督、资源；闫忠民：方法学、正式分析、审阅与编辑稿撰写；余国贤：监督、调查、方法学、正式分析、审阅与编辑稿撰写。

数据可用性

我们的代码和收集的数据集已在 <https://www.sdu-idea.cn/codes.php?name=HerbToxNet> 分享。

利益冲突声明

余国贤报告称，本研究得到了中华人民共和国科学技术部的资助。如果有其他作者，他们声明没有已知的可能影响本报告工作的竞争性经济利益或个人关系。

致谢

本研究得到了中国国家重点研发计划（编号 2022YFC3502101）、中国国家自然科学基金（编号

62272276，）以及山东省自然科学基金（编号 ZR2024JQ001）以及泰山学者计划（编号 tsqn202306007，tsqn202408317）的资助。

补充材料

与本文相关的补充材料可在在线版本中找到，网址为 [10.1016/j.eswa.2025.129969](https://doi.org/10.1016/j.eswa.2025.129969)。

参考文献

- 凯撒, L. & Cech N. (2019). Synergy and antagonism in natural product extracts: when 1 + 1 does not equal 2. *Natural Product Reports* 36(6), 869–888. <https://doi.org/10.1039/C9NP00011A>.
- K. (2003). Some aspects of toxic contaminants in herbal medicines. *Chemosphere* 52(9), 1361–1371. [https://doi.org/10.1016/S0045-6535\(03\)00471-5](https://doi.org/10.1016/S0045-6535(03)00471-5).
- M., & Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *Int. conf. mach. learn.* (pp. 1597–1607). <https://doi.org/10.5555/3524938.3525087>.
- A. Muratov E. Fourches D. Varnek A. Baskin I. Cronin M. Dearden J. Gramatica P. Martin Y. Todeschini R. et al. (2014). Qsar modeling: Where have you been? where are you going to? *Journal of Medicinal Chemistry* 57(12), 4977–5010. <https://doi.org/10.1021/jm4004285>.
- S. Dong L. Liu L. Guo J. Zhao L. Zhang J. Bu D. Liu X. Huo P. Cao W. et al. (2021). Herb: A high-throughput experiment-and reference-guided database of traditional chinese medicine. *Nucleic Acids Research* 49(D1), D1197–D1206. <https://doi.org/10.1093/nar/gkaa1063>.
- S. Shrivastava R. & Mittal M. (2013). Chemistry and pharmacological properties of some natural and synthetic antioxidants for heavy metal toxicity. *Current Medicinal Chemistry* 20(36), 4540–4574. <https://doi.org/10.2174/09298673113209990146>.
- Zou, J., Yan, B., Liu, K., He, S., & Bo, X. (2025). Multi-task multi-view and iterative error-correcting random forest for acute toxicity prediction. *Expert Systems with Applications* 274, 126972. <https://doi.org/10.1016/j.eswa.2025.126972>.
- Y. Liang A. Fan X. Hu L. Hao F. & Li Y. (2019). Safety research in traditional chinese medicine: methods applications and outlook. *Engineering* 5(1), 76–82. <https://doi.org/10.1016/j.eng.2018.11.019>.
- 葛, Z. Jiang X. Tong Z. Feng P. Zhou B. Xu M. Wang Z. & Pang Y. (2021). Multi-label correlation guided feature fusion network for abnormal ECG diagnosis. *Knowledge-Based System* 233, 107508. <https://doi.org/10.1016/j.knosys.2021.107508>.
- L. Jiang J. Chen S. & Qi M. (2023). A syndrome differentiation model of TCM based on multi-label deep forest using biomedical text mining. *Frontiers in Genetics* 14, 1272016. <https://doi.org/10.3389/fgene.2023.1272016>.
- Y. (2006). Dimensionality reduction by learning an invariant mapping. In *IEEE conf. comput. vis. pattern recognit. (cvpr)* (pp. 1735–1742). <https://doi.org/10.1109/CVPR.2006.100>.
- 哈德塞尔, R., Chopra, S., & LeCun,

further indicates that the node itself plays a crucial role in generating its own representation during the learning process.

For HIH meta-path, *Chai Hu* is connected to *Zi Xiao Rong*, *Jian Xue Feng Hou*, and *Ku Lian Pi*. *Chai Hu*, *Zi Xiao Rong*, *Ku Lian Pi*, and *Jian Xue Feng Hou* share common classes of secondary metabolites, which include triterpenoid saponins, flavonoids, phenolic compounds, volatile oils, and bitter alkaloids. For instance, *Chai Hu* contains saikosaponins, *Zi Xiao Rong* is rich in flavonoids and phenolic acids, *Ku Lian Pi* contains toosendanin and related limonoids, while *Jian Xue Feng Hou* harbors cardiotoxic cardiac glycosides such as antiarin. Despite their structural diversity, these ingredients share overlapped biological activities, notably anti-inflammatory, antitumor, and detoxifying effects.

For the HTH meta-path, *Chai Hu* is connected to *Bai Fu Zi*, *Song Luo*, and *Mang Cao*. These herbs share several potential pharmacological targets with *Chai Hu*, including *TNF*, *IL6*, *PTGS2*, and *MAPK1*, which are closely associated with liver injury and inflammatory responses. Saikosaponins in *Chai Hu* and diterpenoid alkaloids in *Bai Fu Zi* may upregulate pro-inflammatory cytokines such as *TNF* and *IL6* via activation of the *NFKB* signaling pathway. Usnic acid in *Song Luo* has been implicated in mitochondrial dysfunction and oxidative stress, leading to hepatotoxicity mediated by *PTGS2* overexpression. Alkaloids in *Mang Cao* may affect *MAPK1* signaling, which is involved in hepatocyte apoptosis and stress response.

In summary, the node-level attention mechanism highlights key associations based on shared herb properties, ingredients, and targets, offering biologically meaningful and interpretable insights into the model’s predictions for *Chai Hu*.

4.7.2. Biointerpretability analysis

To enhance the biological interpretability of HerbToxNet, we constructed a multi-level explanatory framework along the “herb-ingredient-target-biological pathway-toxicity” axis. Taking *Chai Hu* as a representative case, we systematically identified its toxic ingredients, related molecular targets, and associated toxicity types by integrating meta-path analysis, literature curation, and toxicity databases. We further linked these components via their biological pathways and functional annotations. Our results show that HerbToxNet successfully uncovers key toxic ingredients and their molecular targets involved in hepatotoxicity, nephrotoxicity, and systemic toxicity, and reveals the relevant biological pathways and mechanisms underlying herb-induced toxic responses. This demonstrates that HerbToxNet can provide mechanistically meaningful and biologically plausible explanations for its toxicity predictions, thereby supporting more reliable risk assessment of herbs. A comprehensive description of the analytical procedures, in-depth results, and all supporting evidence can be found in Supplementary Section V.

5. Conclusion

In this paper, we construct a multi-label herb toxicity dataset and propose HerbToxNet, a framework for herb toxicity prediction that integrates heterogeneous network modeling, contrastive learning, and weighted label fusion. Empirical study shows that HerbToxNet achieves superior performance than competitive methods. HerbToxNet can identify novel potential toxicities and improve the safe clinical use of herbs. Moreover, it can uncover toxicity-related ingredients and targets, provide insights into toxicity mechanisms, guide antidote development, support toxicity-related databases construction and the discovery of safer herbs. Although HerbToxNet achieves better performance than compared methods and gives some authentic results, it still has limitations. Our dataset currently covers five toxicity categories, with many herbs assigned multiple labels, supporting the use of multi-label classification. Due to limited standardized data, broader toxicity categories are not included at this stage. In future work, we will expand toxicity coverage to enhance the robustness and applicability of HerbToxNet.

CRedit authorship contribution statement

Yongzheng Zhu: Data curation, Formal analysis, Investigation, Software, Validation, Writing – original draft; **Yunbo Miao:** Formal analysis, Software, Validation, Writing – original draft; **Rong Sun:** Supervision, Resources; **Zhongmin Yan:** Methodology, Formal analysis, Writing – review & editing; **Guoxian Yu:** Supervision, Investigation, Methodology, Formal analysis, Writing – review & editing.

Data Availability

Our code and collected dataset are shared at <https://www.sdu-idea.cn/codes.php?name=HerbToxNet>.

Declaration of competing interest

Guoxian Yu reports financial support was provided by Ministry of Science and Technology of the People’s Republic of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by the [National Key Research and Development Program of China](#) [grant numbers 2022YFC3502101]; [National Natural Science Foundation of China](#) [grant numbers 62272276, 62432006, [Natural Science Foundation of Shandong Province](#) [grant numbers ZR2024JQ001] and [Taishan Scholars Program](#) [grant numbers tsqn202306007, tsqn202408317].

Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.eswa.2025.129969](https://doi.org/10.1016/j.eswa.2025.129969).

References

- Caesar, L., & Cech, N. (2019). Synergy and antagonism in natural product extracts: when 1 + 1 does not equal 2. *Natural Product Reports*, 36(6), 869–888. <https://doi.org/10.1039/C9NP00011A>.
- Chan, K. (2003). Some aspects of toxic contaminants in herbal medicines. *Chemosphere*, 52(9), 1361–1371. [https://doi.org/10.1016/S0045-6535\(03\)00471-5](https://doi.org/10.1016/S0045-6535(03)00471-5).
- Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *Int. conf. mach. learn.* (pp. 1597–1607). <https://doi.org/10.5555/3524938.3525087>.
- Cherkasov, A., Muratov, E., Fourches, D., Varnek, A., Baskin, I., Cronin, M., Dearden, J., Gramatica, P., Martin, Y., Todeschini, R. et al. (2014). Qsar modeling: Where have you been? where are you going to? *Journal of Medicinal Chemistry*, 57(12), 4977–5010. <https://doi.org/10.1021/jm4004285>.
- Fang, S., Dong, L., Liu, L., Guo, J., Zhao, L., Zhang, J., Bu, D., Liu, X., Huo, P., Cao, W. et al. (2021). Herb: A high-throughput experiment-and reference-guided database of traditional chinese medicine. *Nucleic Acids Research*, 49(D1), D1197–D1206. <https://doi.org/10.1093/nar/gkaa1063>.
- Flora, S., Shrivastava, R., & Mittal, M. (2013). Chemistry and pharmacological properties of some natural and synthetic antioxidants for heavy metal toxicity. *Current Medicinal Chemistry*, 20(36), 4540–4574. <https://doi.org/10.2174/09298673113209990146>.
- Gao, J., Wu, L., Lin, G., Zou, J., Yan, B., Liu, K., He, S., & Bo, X. (2025). Multi-task multi-view and iterative error-correcting random forest for acute toxicity prediction. *Expert Systems with Applications*, 274, 126972. <https://doi.org/10.1016/j.eswa.2025.126972>.
- Gao, Y., Liang, A., Fan, X., Hu, L., Hao, F., & Li, Y. (2019). Safety research in traditional chinese medicine: methods, applications, and outlook. *Engineering*, 5(1), 76–82. <https://doi.org/10.1016/j.eng.2018.11.019>.
- Ge, Z., Jiang, X., Tong, Z., Feng, P., Zhou, B., Xu, M., Wang, Z., & Pang, Y. (2021). Multi-label correlation guided feature fusion network for abnormal ECG diagnosis. *Knowledge-Based System*, 233, 107508. <https://doi.org/10.1016/j.knosys.2021.107508>.
- Gong, L., Jiang, J., Chen, S., & Qi, M. (2023). A syndrome differentiation model of TCM based on multi-label deep forest using biomedical text mining. *Frontiers in Genetics*, 14, 1272016. <https://doi.org/10.3389/fgene.2023.1272016>.
- Hadsell, R., Chopra, S., & LeCun, Y. (2006). Dimensionality reduction by learning an invariant mapping. In *IEEE conf. comput. vis. pattern recognit. (cvpr)* (pp. 1735–1742). <https://doi.org/10.1109/CVPR.2006.100>.

他, S., Zhang, C., Zhou, P., Zhang, X., Ye, T., Wang, R., Sun, G., & Sun, X. (2019a). Herb-induced liver injury: Phylogenetic relationship structure-toxicity relationship and herb-ingredient network analysis. *International Journal of Molecular Sciences* 20(15) 3633. https://doi.org/10.3390/ijms20153633.

S. Zhang X. Lu S. Zhu T. Sun G. & Sun X. (2019b). A computational toxicology approach to screen the hepatotoxic ingredients in traditional chinese medicines: Polygonum multiflorum thunb as a case study. *Biomolecules* 9(10) 577. https://doi.org/10.3390/103390_9100577

H. Li Y. Zheng Z. Hu W. Lin R. & Kang Y. (2025). A traditional chinese medicine prescription recommendation model based on contrastive pre-training and hierarchical structure network. *Expert Systems with Applications* 268 126318. https://doi.org/10.1016/j.eswa.2024.126318.

C. Li X. Hu S. Liu G. Fang J. Zhou X. Yan X. & Yan B. (2024). Advanced mass-spectra-based machine learning for predicting the toxicity of traditional chinese medicines. *Analytical Chemistry*, 97(1), 783–792. https://doi.org/10.1021/acs.analchem.4c05311.

江, M. Yang J. Zhang C. Liu B. Chan K. Cao H. & Lu A. (2010). Clinical studies with traditional chinese medicine in the past decade and future research and development. *Planta Medica*, 76(17), 2048–2064. https://doi.org/10.1055/s-0030-1250456.

A. Trifan A. Vieriu M. Caba I.-C. Mărtu I. & Agoroaei L. (2022). Current trends in toxicity assessment of herbal medicines: A narrative review. *Processes* 11(1) 83. https://doi.org/10.3390/pr11010083.

P. Teterwak P. Wang C. Sarna A. Tian Y. Isola P. Maschinot A. Liu C. & Krishnan D. (2020). Supervised contrastive learning. In *Adv. neural inf. process. syst.* (pp. 18661–18673). https://doi.org/10.5555/3495724.3497291.

Roberts, R., Liu, Z., & Thakkar, S. (2020). DeepDILI: Deep learning-driven drug-induced liver injury prediction using model-level representation. *Chemical Research in Toxicology* 34(2) 550–565. https://doi.org/10.1021/acs.chemrestox.0c00374

Y. Zhang Y. Wang Y. Li Y. Yang F. Zhang P. Zhang Y. & Liu C. (2019). A strategy for the discovery and validation of toxicity quality marker of chinese medicine based on network toxicology. *Phytomedicine* 54 365–370. https://doi.org/10.1016/j.phymed.2018.01.018.

Li X. Huang N. Fan M. & Sun R. (2020). Toxicity of traditional Chinese medicine herbal and mineral products. *Advances in Pharmacology* 87 301–346. https://doi.org/10.1016/bs.apha.2019.08.001.

Q. Chen G. He H. Yang Z. Zhao L. Zhang K. & Chen C.Y.C. (2023). Tcmbank-the largest tcm database provides deep learning-based chinese-western medicine exclusion prediction. *Signal Transduction and Targeted Therapy* 8(1) 127. https://doi.org/10.1016/s41392-023-01339-1

范, T., Tao, X., 127. https://doi.org/10.1016/s41392-023-01339-1

Zhang, J., Yong, J., Li, Y., & Xie, H. (2022). Graph-based multi-label disease prediction model learning from medical data and domain knowledge. *Knowledge-Based Systems*, 235, 107662. https://doi.org/10.1016/j.knosys.2021.107662.

Y. Zhang Y. Peng S. Huo M. Li J. Cao Y. & Chen Z. (2022). Property theory of chinese materia medica: Clinical pharmacodynamics of traditional chinese medicine. *Journal of Traditional Chinese Medical Sciences* 9(1) 7–12. https://doi.org/10.1016/j.jtcms.2022.01.006.

S. Wang M. Yang Y. Yu G. Wang J. Yan Z. Domeniconi C. & Guo M. (2023). Meta multi-instance multi-label learning by heterogeneous network fusion. *An International Journal on Information Fusion* 94 272–283. https://doi.org/10.1016/j.inffus.2023.02.010

A., & Bajic, V. (2016). In silico toxicology: Computational methods for the prediction of chemical toxicity. *Wiley Interdisciplinary Reviews Computational Molecular Science* 6(2) 147–172. https://doi.org/10.1002/wcms.1240.

T. Ben-Baruch E. Zamir N. Noy A. Friedman I. Protter M. & Zelnik-Manor L. (2021). Asymmetric loss for multi-label classification. In *IEEE/CVF int. conf. comput. vis.* (pp. 82–91). https://doi.org/10.1109/ICCV48922.2021.00015.

Sun, Y. Shi S. Li Y. & Wang Q. (2019). Development of quantitative structure-activity relationship models to predict potential nephrotoxic ingredients in traditional chinese medicines. *Food and Chemical Toxicology* 128 163–170. https://doi.org/10.1016/j.fct.2019.03.056.

Wang, X. Ji H. Shi C. Wang B. Ye Y. Cui P. & Yu P. (2019). Heterogeneous graph attention network. In *Www conf.* (pp. 2022–2032). https://doi.org/10.1145/3308558.3313562

Wang, X. Qu H. Liu P. & Cheng Y. (2004). A self-learning expert system for diagnosis in traditional chinese medicine. *Expert Systems with Applications* 26(4) 557–566. https://doi.org/10.1016/j.eswa.2003.10.004.

Z. Li L. Yan J. & Yao Y. (2020). Approaching high-accuracy side effect prediction of traditional chinese medicine compound prescription using network embedding and deep learning. *IEEE Access* 8 82493–82499. https://doi.org/10.1109/ACCESS.2020.2991750

J. Dai J. Sun Y. Meng Z. Ma H. & Zhou Y. (2024). Tirpnet: Risk prediction of traditional chinese medicine ingredients based on a deep neural network. *Journal of Ethnopharmacology* 325 117860. https://doi.org/10.1016/j.jep.2024.117860.

Wu, Q. Cai C. Guo P. Chen M. Wu X. Zhou J. Luo Y. Zou Y. Liu A. Wang Q. et al. (2019a). In silico identification and mechanism exploration of hepatotoxic ingredients in traditional chinese medicine. *Frontiers in Pharmacology* 10 458. https://doi.org/10.3389/fphar.2019.00458.

Q. Yin C. Li Y. Cai J. Yang H. Huang Y. Zheng Y. Xiong K. Yu H. Lu A. et al. (2021). Detecting critical functional ingredients group and mechanism of xuebijing injection in treating sepsis. *Frontiers in Pharmacology* 12 3389/fphar.2021.769190.

Y. Zhang F. Yang K. Fang S. Bu D. Li H. Sun L. Hu H. Gao K. Wang W. et al. (2019b). SymMap: An integrative database of traditional chinese medicine enhanced by symptom mapping. *Nucleic Acids Research* 47(D1) 3389–2021 769190.

X. Xu H. Shang Y. Zhu R. Hong X. Song Z. & Yang Z. (2021). Development of the general chapters of the chinese pharmacopoeia 2020 edition: A review. *Journal of Pharmaceutical Analysis* 11(4) 398–404. https://doi.org/10.1016/j.jpba.2021.05.001

J., Wu, L., He, S., Liu, K., & Bo, X. (2022a). An algorithm framework for drug-induced liver injury prediction based on genetic algorithm and ensemble learning. *Molecules*, 27(10), 3112. https://doi.org/10.3390/molecules27103112.

D. Zheng G. Wang C. Chen Z. Mao T. Gao J. Yan Y. Chen X. Ji X. Yu J. Yang, et al. (2022b). Hit 2.0: An enhanced platform for herbal ingredients' targets. *Nucleic Acids Research* 50(D1) D1238–D1243. https://doi.org/10.1093/nar/gkab1011.

J. Wang S. Zhang T. Sun Y. Han L. Banahene P. O. & Wang Q. (2021). Predicting the potential toxicity of 26 components in cassiae semen using in silico and in vitro approaches. *Current Research in Toxicology* 2 237–245. https://doi.org/10.1016/j.crttox.2021.06.001

K. Zhou X. Zhang R. Liu B. Lei L. Zhang X. Chu H. Sun C. Gao Z. & Xu Yu, H. (2015). Integrating herb effect similarity for network-based herb target prediction. In *Int. conf. biomed. eng. inform.* (pp. 483–488). https://doi.org/10.1109/BMEI.2015.7401553

Z. Kou F. Gao Y. Gao F. Lyu C.-m. & Wei H. (2025). A machine learning model for predicting abnormal liver function induced by a Chinese herbal medicine preparation (Zhengqing fengtongning) in patients with rheumatoid arthritis based on real-world study. *Journal of Integrative Medicine* 23(1) 25–35. https://doi.org/10.1016/j.joim.2024.12.001

Y. Ren L. Sun R. Wang J. & Yu G. (2025). Herb-target interaction prediction by multi-instance learning. *IEEE Transactions on Artificial Intelligence* 6(5) 1184–1193. https://doi.org/10.1109/TAI.2024.3515932.

He, S., Zhang, C., Zhou, P., Zhang, X., Ye, T., Wang, R., Sun, G., & Sun, X. (2019a). Herb-induced liver injury: Phylogenetic relationship, structure-toxicity relationship, and herb-ingredient network analysis. *International Journal of Molecular Sciences*, 20(15), 3633. https://doi.org/10.3390/ijms20153633.

He, S., Zhang, X., Lu, S., Zhu, T., Sun, G., & Sun, X. (2019b). A computational toxicology approach to screen the hepatotoxic ingredients in traditional chinese medicines: Polygonum multiflorum thunb as a case study. *Biomolecules*, 9(10), 577. https://doi.org/10.3390/biom9100577.

Hu, H., Li, Y., Zheng, Z., Hu, W., Lin, R., & Kang, Y. (2025). A traditional chinese medicine prescription recommendation model based on contrastive pre-training and hierarchical structure network. *Expert Systems with Applications*, 268, 126318. https://doi.org/10.1016/j.eswa.2024.126318.

Jia, C., Li, X., Hu, S., Liu, G., Fang, J., Zhou, X., Yan, X., & Yan, B. (2024). Advanced mass-spectra-based machine learning for predicting the toxicity of traditional chinese medicines. *Analytical Chemistry*, 97(1), 783–792. https://doi.org/10.1021/acs.analchem.4c05311.

Jiang, M., Yang, J., Zhang, C., Liu, B., Chan, K., Cao, H., & Lu, A. (2010). Clinical studies with traditional chinese medicine in the past decade and future research and development. *Planta Medica*, 76(17), 2048–2064. https://doi.org/10.1055/s-0030-1250456.

Jităreanu, A., Trifan, A., Vieriu, M., Caba, I.-C., Mărtu, I., & Agoroaei, L. (2022). Current trends in toxicity assessment of herbal medicines: A narrative review. *Processes*, 11(1), 83. https://doi.org/10.3390/pr11010083.

Khosla, P., Teterwak, P., Wang, C., Sarna, A., Tian, Y., Isola, P., Maschinot, A., Liu, C., & Krishnan, D. (2020). Supervised contrastive learning. In *Adv. neural inf. process. syst.* (pp. 18661–18673). https://doi.org/10.5555/3495724.3497291.

Li, T., Tong, W., Roberts, R., Liu, Z., & Thakkar, S. (2020). DeepDILI: Deep learning-powered drug-induced liver injury prediction using model-level representation. *Chemical Research in Toxicology*, 34(2), 550–565. https://doi.org/10.1021/acs.chemrestox.0c00374.

Li, Y., Zhang, Y., Wang, Y., Li, Y., Yang, F., Zhang, P., Zhang, Y., & Liu, C. (2019). A strategy for the discovery and validation of toxicity quality marker of chinese medicine based on network toxicology. *Phytomedicine*, 54, 365–370. https://doi.org/10.1016/j.phymed.2018.01.018.

Liu, R., Li, X., Huang, N., Fan, M., & Sun, R. (2020). Toxicity of traditional Chinese medicine herbal and mineral products. *Advances in Pharmacology*, 87, 301–346. https://doi.org/10.1016/bs.apha.2019.08.001.

Lv, Q., Chen, G., He, H., Yang, Z., Zhao, L., Zhang, K., & Chen, C. Y. C. (2023). Tcmbank-the largest tcm database provides deep learning-based chinese-western medicine exclusion prediction. *Signal Transduction and Targeted Therapy*, 8(1), 127. https://doi.org/10.1038/s41392-023-01339-1.

Pham, T., Tao, X., Zhang, J., Yong, J., Li, Y., & Xie, H. (2022). Graph-based multi-label disease prediction model learning from medical data and domain knowledge. *Knowledge-Based System*, 235, 107662. https://doi.org/10.1016/j.knosys.2021.107662.

Qiao, Y., Zhang, Y., Peng, S., Huo, M., Li, J., Cao, Y., & Chen, Z. (2022). Property theory of chinese materia medica: Clinical pharmacodynamics of traditional chinese medicine. *Journal of Traditional Chinese Medical Sciences*, 9(1), 7–12. https://doi.org/10.1016/j.jtcms.2022.01.006.

Qiu, S., Wang, M., Yang, Y., Yu, G., Wang, J., Yan, Z., Domeniconi, C., & Guo, M. (2023). Meta multi-instance multi-label learning by heterogeneous network fusion. *An International Journal on Information Fusion*, 94, 272–283. https://doi.org/10.1016/j.inffus.2023.02.010.

Raies, A., & Bajic, V. (2016). In silico toxicology: Computational methods for the prediction of chemical toxicity. *Wiley Interdisciplinary Reviews Computational Molecular Science*, 6(2), 147–172. https://doi.org/10.1002/wcms.1240.

Ridnik, T., Ben-Baruch, E., Zamir, N., Noy, A., Friedman, I., Protter, M., & Zelnik-Manor, L. (2021). Asymmetric loss for multi-label classification. In *IEEE/CVF int. conf. comput. vis.* (pp. 82–91). https://doi.org/10.1109/ICCV48922.2021.00015.

Silberg, J., Swanson, K., Simon, E., Zhang, A., Ghazizadeh, Z., Ogden, S., Hamadeh, H., & Zou, J. (2024). Unitox: Leveraging LLMs to curate a unified dataset of drug-induced toxicity from FDA labels. *Advances in Neural Information Processing Systems*, 37, 12078–12093. https://doi.org/10.1101/2024.06.21.24309315.

Sun, Y., Shi, S., Li, Y., & Wang, Q. (2019). Development of quantitative structure-activity relationship models to predict potential nephrotoxic ingredients in traditional chinese medicines. *Food and Chemical Toxicology*, 128, 163–170. https://doi.org/10.1016/j.fct.2019.03.056.

Wang, X., Ji, H., Shi, C., Wang, B., Ye, Y., Cui, P., & Yu, P. (2019). Heterogeneous graph attention network. In *Www conf.* (pp. 2022–2032). https://doi.org/10.1145/3308558.3313562.

Wang, X., Qu, H., Liu, P., & Cheng, Y. (2004). A self-learning expert system for diagnosis in traditional chinese medicine. *Expert Systems with Applications*, 26(4), 557–566. https://doi.org/10.1016/j.eswa.2003.10.004.

Wang, Z., Li, L., Yan, J., & Yao, Y. (2020). Approaching high-accuracy side effect prediction of traditional chinese medicine compound prescription using network embedding and deep learning. *IEEE Access*, 8, 82493–82499. https://doi.org/10.1109/ACCESS.2020.2991750.

Wei, J., Dai, J., Sun, Y., Meng, Z., Ma, H., & Zhou, Y. (2024). Tirpnet: Risk prediction of traditional chinese medicine ingredients based on a deep neural network. *Journal of Ethnopharmacology*, 325, 117860. https://doi.org/10.1016/j.jep.2024.117860.

Wu, Q., Cai, C., Guo, P., Chen, M., Wu, X., Zhou, J., Luo, Y., Zou, Y., Liu, A., Wang, Q. et al. (2019a). In silico identification and mechanism exploration of hepatotoxic ingredients in traditional chinese medicine. *Frontiers in Pharmacology*, 10, 458. https://doi.org/10.3389/fphar.2019.00458.

Wu, Q., Yin, C., Li, Y., Cai, J., Yang, H., Huang, Y., Zheng, Y., Xiong, K., Yu, H., Lu, A. et al. (2021). Detecting critical functional ingredients group and mechanism of xuebijing injection in treating sepsis. *Frontiers in Pharmacology*, 12, 769190. https://doi.org/10.3389/fphar.2021.769190.

Wu, Y., Zhang, F., Yang, K., Fang, S., Bu, D., Li, H., Sun, L., Hu, H., Gao, K., Wang, W. et al. (2019b). SymMap: An integrative database of traditional chinese medicine enhanced by symptom mapping. *Nucleic Acids Research*, 47(D1), D1110–D1117. https://doi.org/10.1093/nar/gky1021.

Xu, X., Xu, H., Shang, Y., Zhu, R., Hong, X., Song, Z., & Yang, Z. (2021). Development of the general chapters of the chinese pharmacopoeia 2020 edition: A review. *Journal of Pharmaceutical Analysis*, 11(4), 398–404. https://doi.org/10.1016/j.jpba.2021.05.001.

Yan, B., Ye, X., Wang, J., Han, J., Wu, L., He, S., Liu, K., & Bo, X. (2022a). An algorithm framework for drug-induced liver injury prediction based on genetic algorithm and ensemble learning. *Molecules*, 27(10), 3112. https://doi.org/10.3390/molecules27103112.

Yan, D., Zheng, G., Wang, C., Chen, Z., Mao, T., Gao, J., Yan, Y., Chen, X., Ji, X., Yu, J. et al. (2022b). Hit 2.0: An enhanced platform for herbal ingredients' targets. *Nucleic Acids Research*, 50(D1), D1238–D1243. https://doi.org/10.1093/nar/gkab1011.

Yang, J., Wang, S., Zhang, T., Sun, Y., Han, L., Banahene, P. O., & Wang, Q. (2021). Predicting the potential toxicity of 26 components in cassiae semen using in silico and in vitro approaches. *Current Research in Toxicology*, 2, 237–245. https://doi.org/10.1016/j.crttox.2021.06.001.

Yang, K., Zhou, X., Zhang, R., Liu, B., Lei, L., Zhang, X., Chu, H., Sun, C., Gao, Z., & Xu, H. (2015). Integrating herb effect similarity for network-based herb target prediction. In *Int. conf. biomed. eng. inform.* (pp. 483–488). https://doi.org/10.1109/BMEI.2015.7401553.

Yu, Z., Kou, F., Gao, Y., Gao, F., Lyu, C.-m., & Wei, H. (2025). A machine learning model for predicting abnormal liver function induced by a Chinese herbal medicine preparation (Zhengqing fengtongning) in patients with rheumatoid arthritis based on real-world study. *Journal of Integrative Medicine*, 23(1), 25–35. https://doi.org/10.1016/j.joim.2024.12.001.

Zhu, Y., Ren, L., Sun, R., Wang, J., & Yu, G. (2025). Herb-target interaction prediction by multi-instance learning. *IEEE Transactions on Artificial Intelligence*, 6(5), 1184–1193. https://doi.org/10.1109/TAI.2024.3515932.